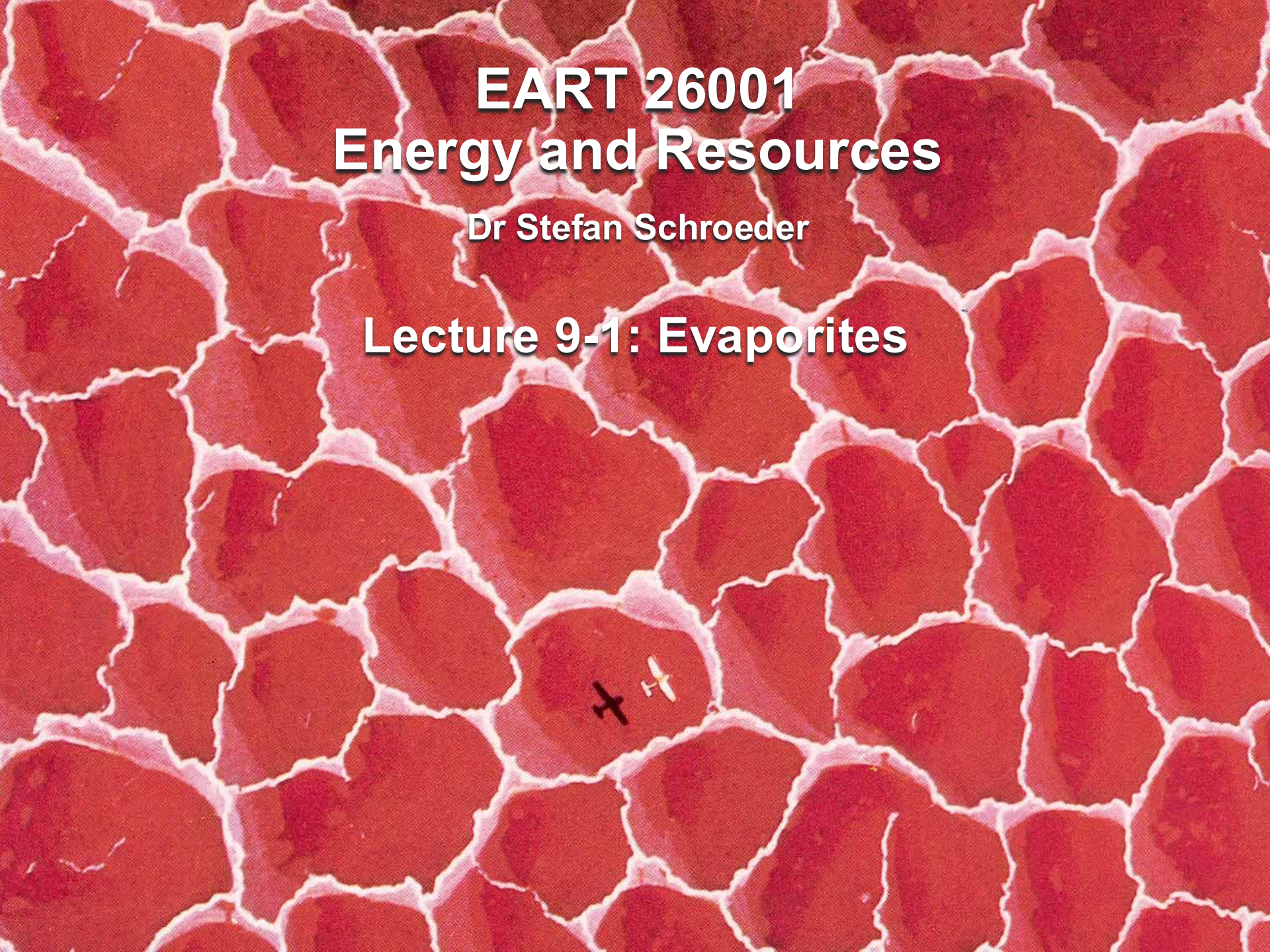


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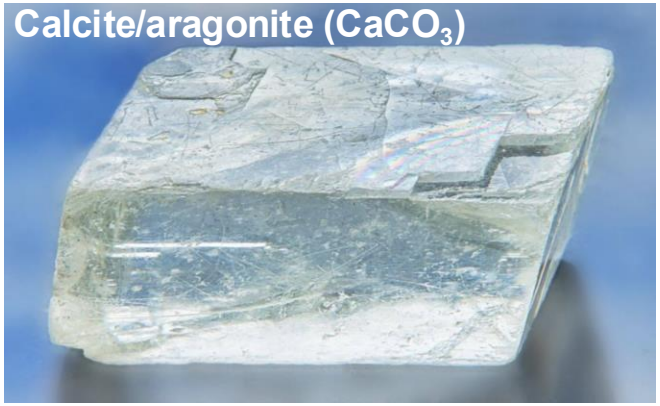
Lecture 9-1: Evaporites

Evaporites
Learning goals week 9

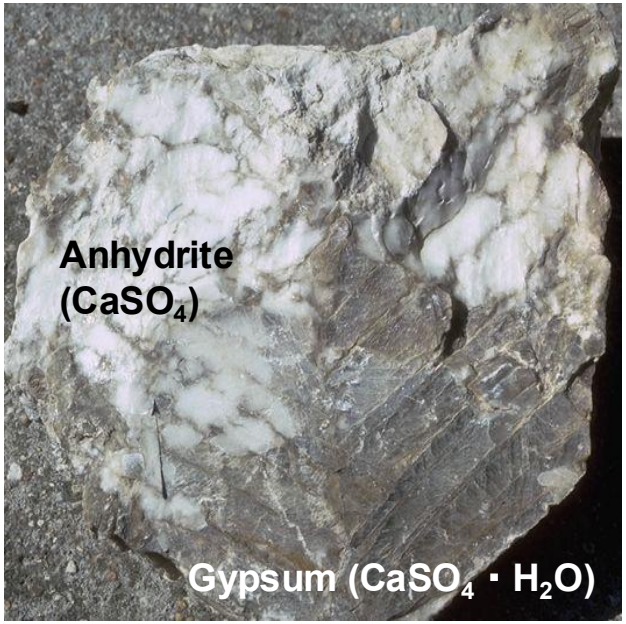
This week you will learn to:

- **Describe the applications of evaporite minerals.**
- **Describe the mechanisms for evaporite formation, and recognize how evaporite mineralogy and fabric relate to depositional environments and processes.**
- **Describe the exploration and production methods for evaporites.**

Evaporites
Typical evaporite minerals

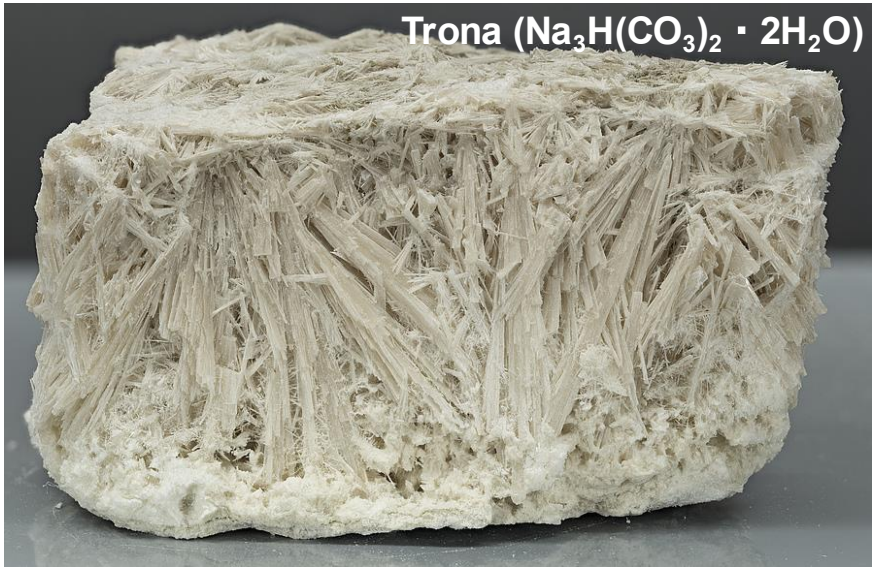


Source: www.mindat.org









Source: www.mindat.org

Evaporites
Introduction

Seawater composition (Mason 1966)

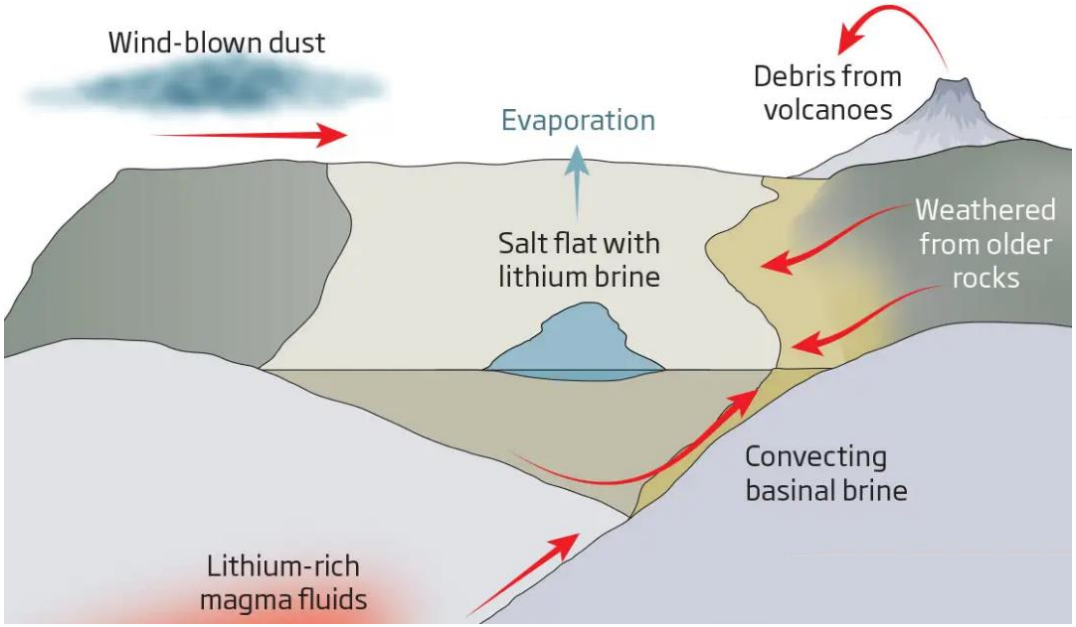
Bicarbonate $(HCO_3)^-$ and carbonate $(CO_3)^{2-}$	0.4
Calcium (Ca^{2+})	1.2
Amorphous silica (H_4SiO_4)	<0.01
Sulfate $(SO_4)^{2-}$	7.7
Chlorine (Cl^-)	55.0
Sodium (Na^+)	30.6
Magnesium (Mg^{2+})	3.7
Potassium (K^+)	1.1
Iron $(Fe^{2+}$ and $Fe^{3+})$	<0.01
Aluminum $Al(OH)_4^-$	<0.01
Nitrate $(NO_3)^-$	<0.01
Total	99.7

typical marine minerals

Tucker (1981)

halite	$NaCl$
sylvite	KCl
carnallite	$KMgCl_3.6H_2O$
kainite	$KMgClSO_4.3H_2O$
anhydrite	$CaSO_4$
gypsum	$CaSO_4.2H_2O$
polyhalite	$K_2MgCa_2(SO_4)_4.2H_2O$
kieserite	$MgSO_4.H_2O$

Ion sources in continental interior basins



Source: USGS

fluid composition 'families'

Ca–Mg–Na–(K)–Cl	Na–(Ca)–SO4–Cl	Mg–Na–(Ca)–SO4–Cl	Na–CO3–Cl	Na–CO3–SO4–Cl

typical non-marine minerals

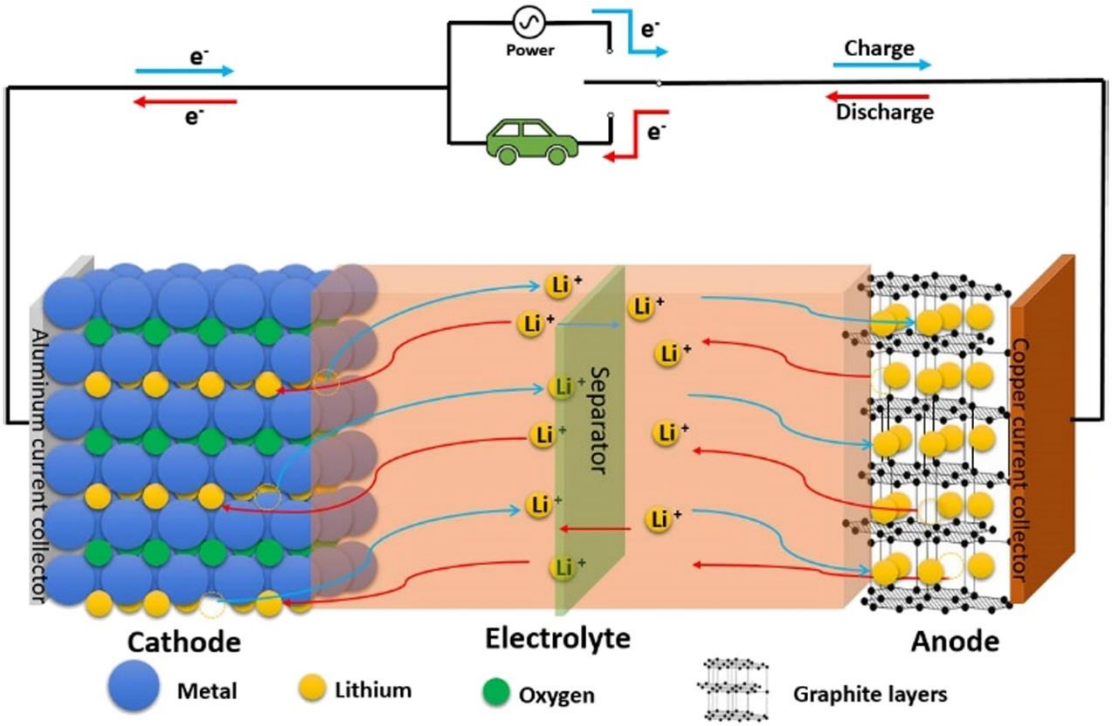
halite, gypsum, anhydrite	
epsomite	$MgSO_4.7H_2O$
trona	$Na_2CO_3.NaHCO_3.2H_2O$
mirabilite	$Na_2SO_4.10H_2O$
thenardite	$NaSO_4$
bloedite	$Na_2SO_4.MgSO_4.4H_2O$
gaylussite	$Na_2CO_3.CaCO_3.5H_2O$
glauberite	$CaSO_4.Na_2SO_4$

Evaporites
Economic use

Type	Source	Uses
Ammonium sulphate	Brine processing	Fertiliser
Borate	Borax (tincal), Boracite	Glass, ceramics, agricultural chemicals, pharmaceuticals, flux cleansing agent, water softener, preservative, fire retardant
Calcium sulphate	Gypsum	Construction industry, plaster, cement retardant, soil enhancer
Iodine	Brine processing	Animal feed supplements, catalysts, ink, colorants, pharmaceutical, photography, sanitary and industrial disinfectant, stabilisers
Lithium	Brine processing, Hectorite in saline clays	Production of ceramics, glass, and primary aluminium, battery technology, pharmaceutical
Potash	Sylvite, Sylvinite, Brine processing	Fertiliser, chemical industry
Sodium carbonate	Trona, Solvay process, Brine processing	Glass manufacture, chemicals, soap and detergents, pulp and paper manufacture, water treatment, and flue gas desulphurisation
Sodium chloride	Halite, Brine processing	Food processing, chemical and industrial products, chloralkali industry, swimming pool additive, stockfeed additive
Sodium sulphate	Mirabilite, Glauberite, Brine processing	Detergent powder, glass, pulp and paper
Strontium	Celestite, Strontianite	Color television tubes, ceramic magnets, coloring agent in pyrotechnics and signals
Sulphur	Biogenic sulphur, Frasch processs	Agricultural chemicals, chemicals manufacture, metal mining, and petroleum refining

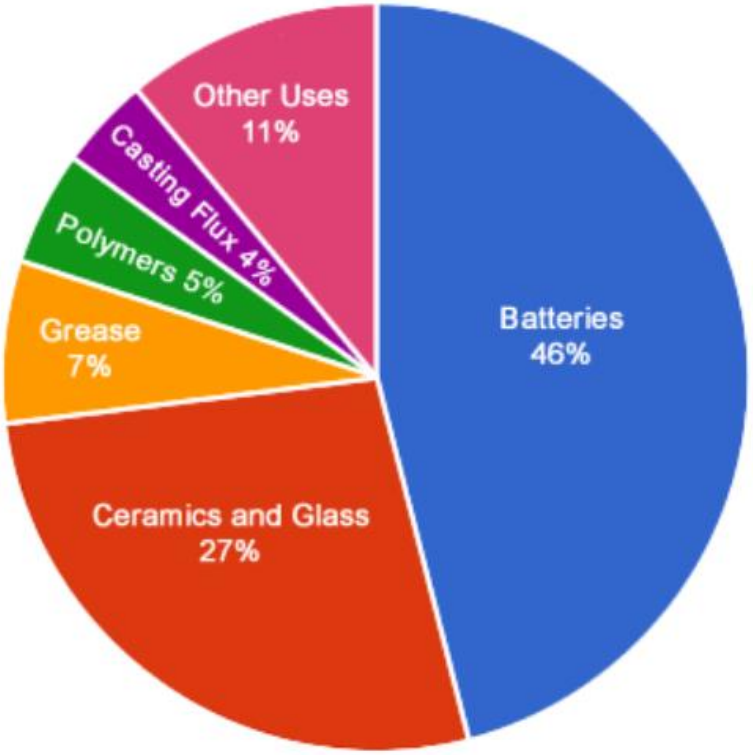
(Warren 1999)

Evaporites
Lithium for batteries



Liang 2018

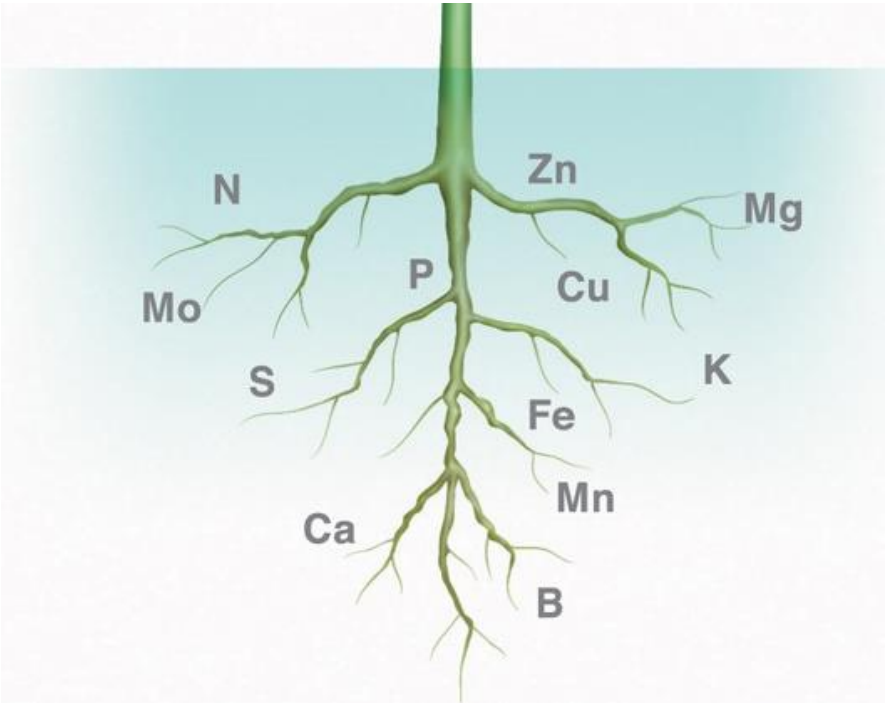
Global use of Li (2018)

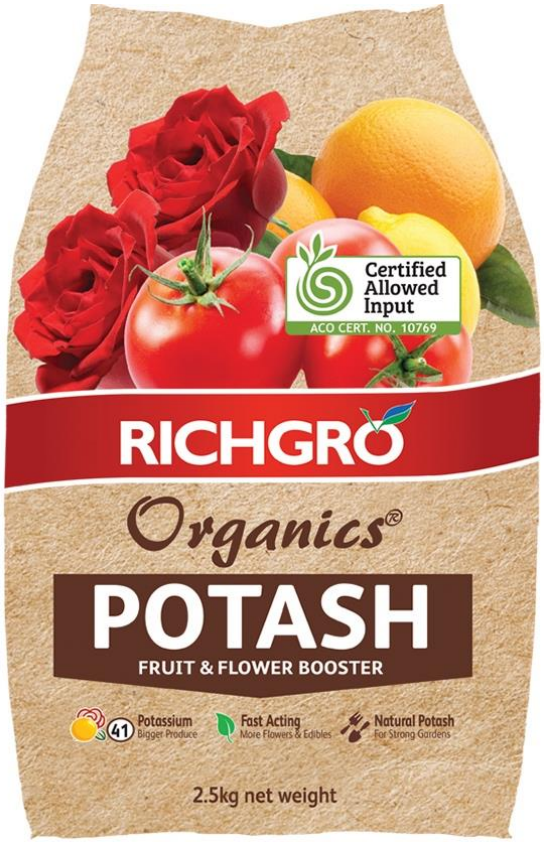


Source: USGS 2018

- Lightest metal
- Highly reactive and flammable (melting point 181°C)
- Concentrations 20ppm in crust and 0.17ppm in sea water

Evaporites
Potassium as fertilizer

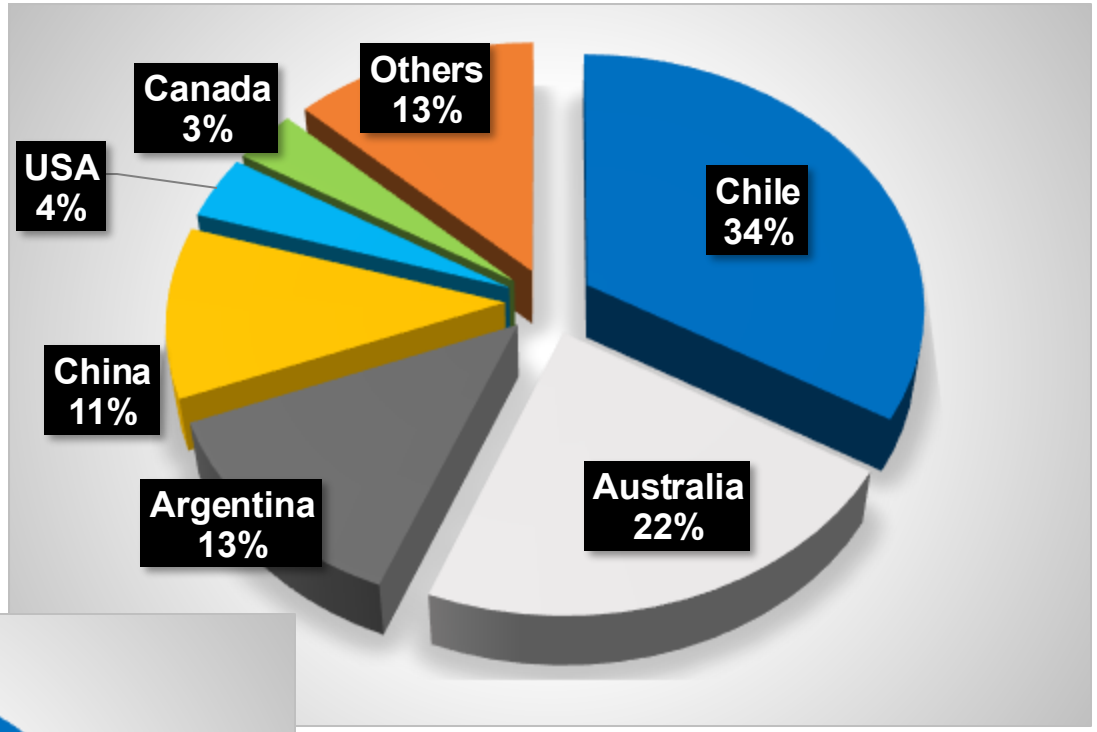




Evaporites
Resources

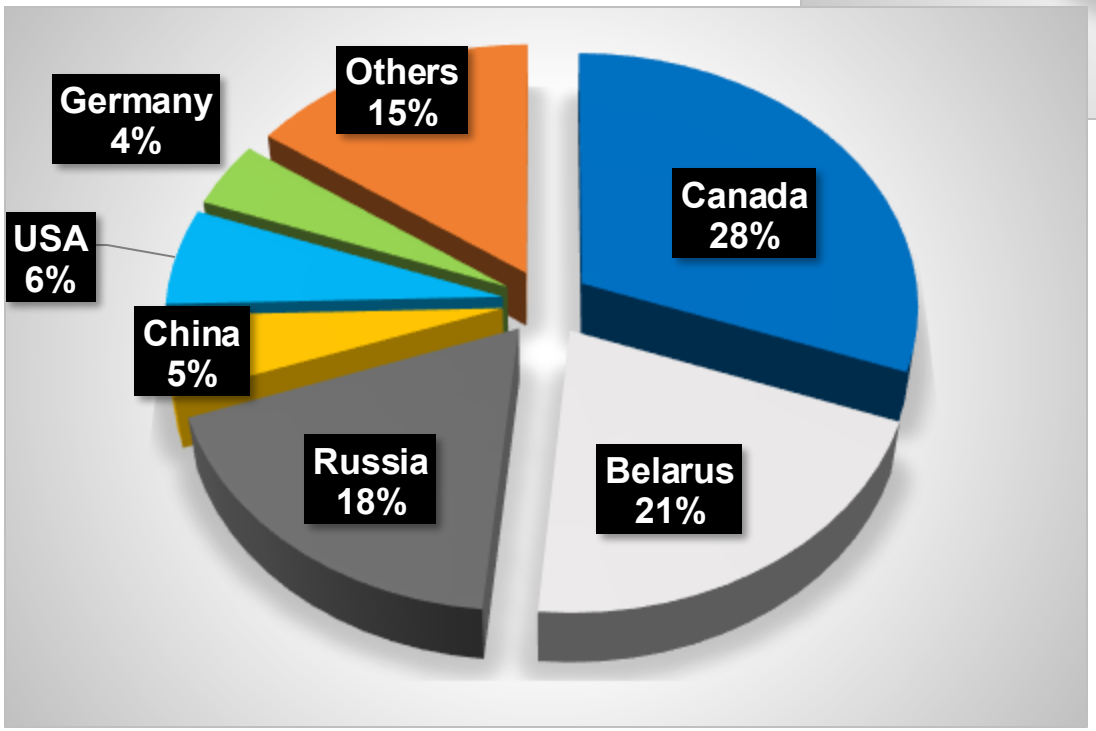
- Supply issues for Li (critical resource), although can be replaced in batteries by Ca, Mg, Hg, Zn
- No substitutes for K

Li reserves 2023



Source: USGS

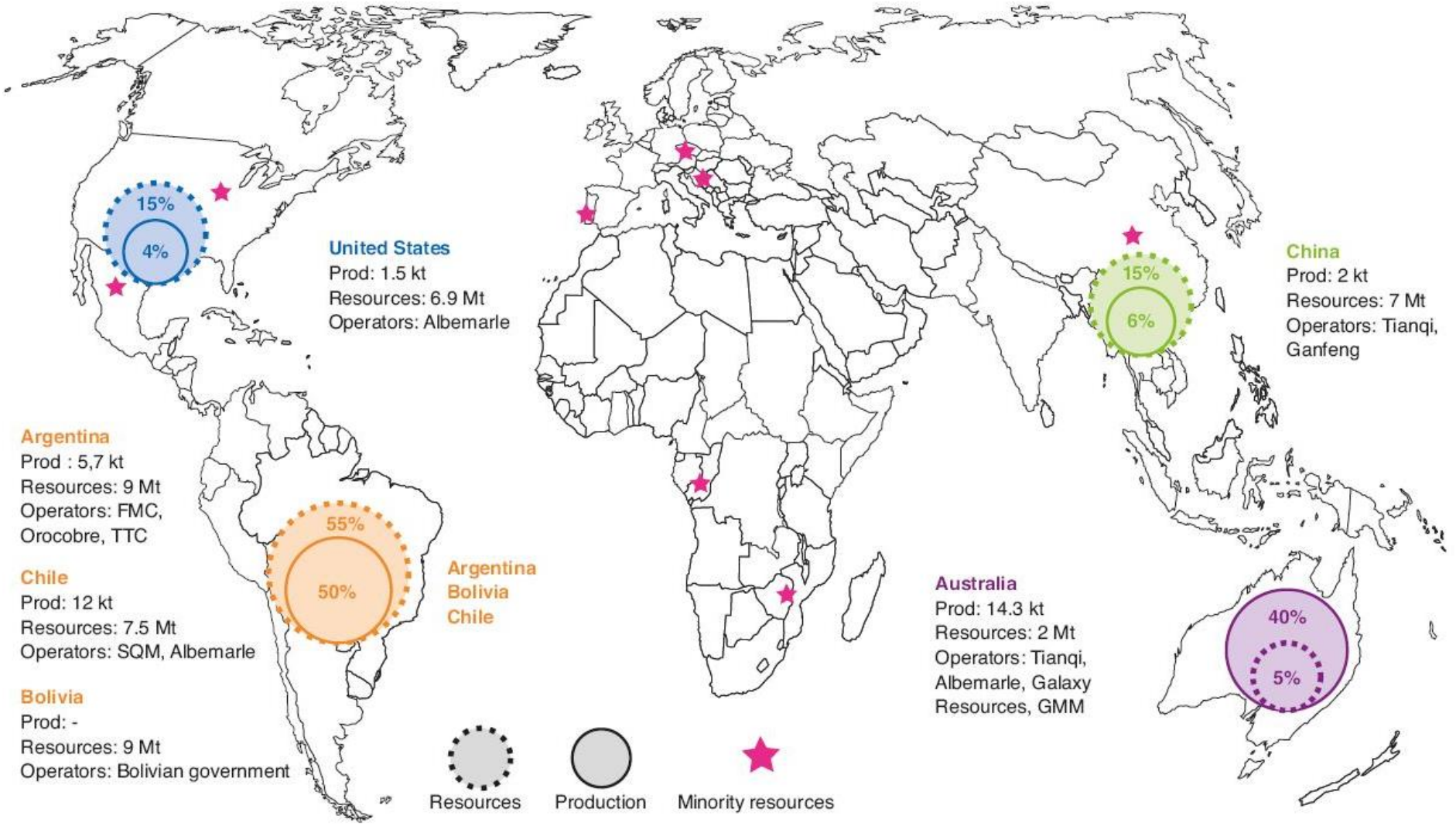
K reserves 2023



Source: USGS

Evaporites
Resources

Li production 2018



Source: USGS, IFP

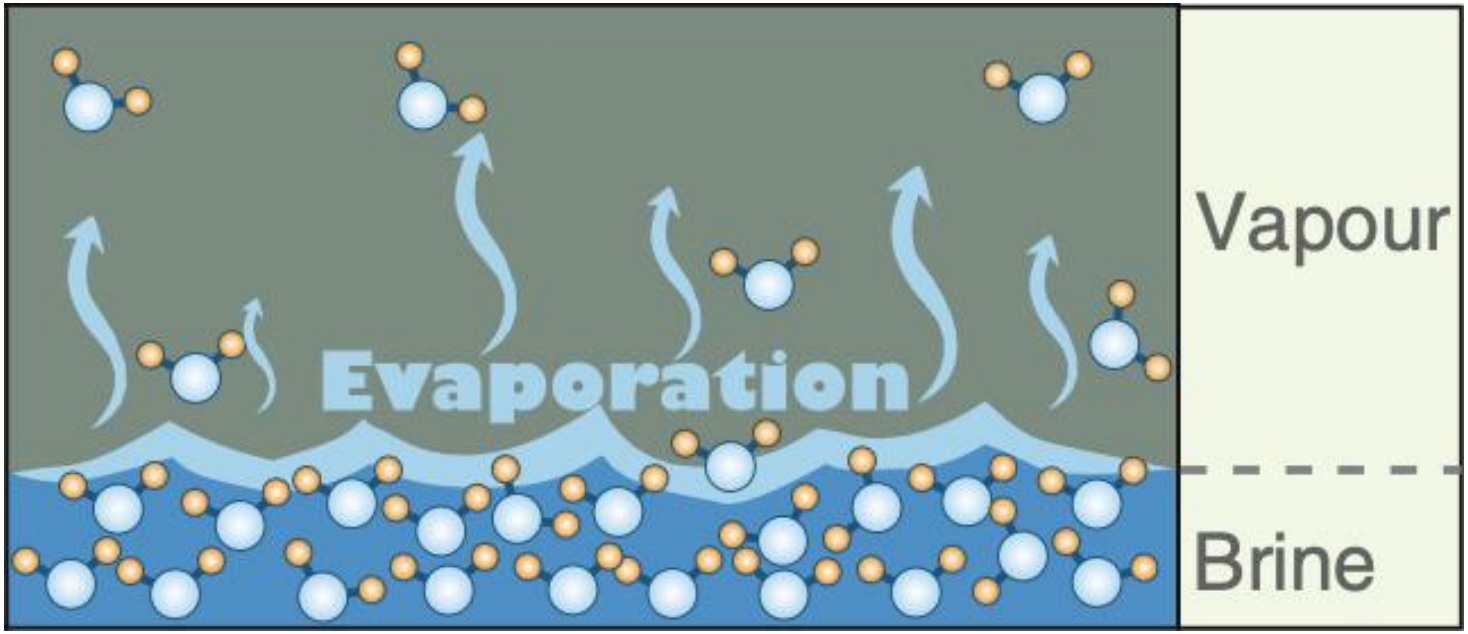
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Lecture 9-2: Evaporite formation and environments



Evaporite formation
Brine evaporation

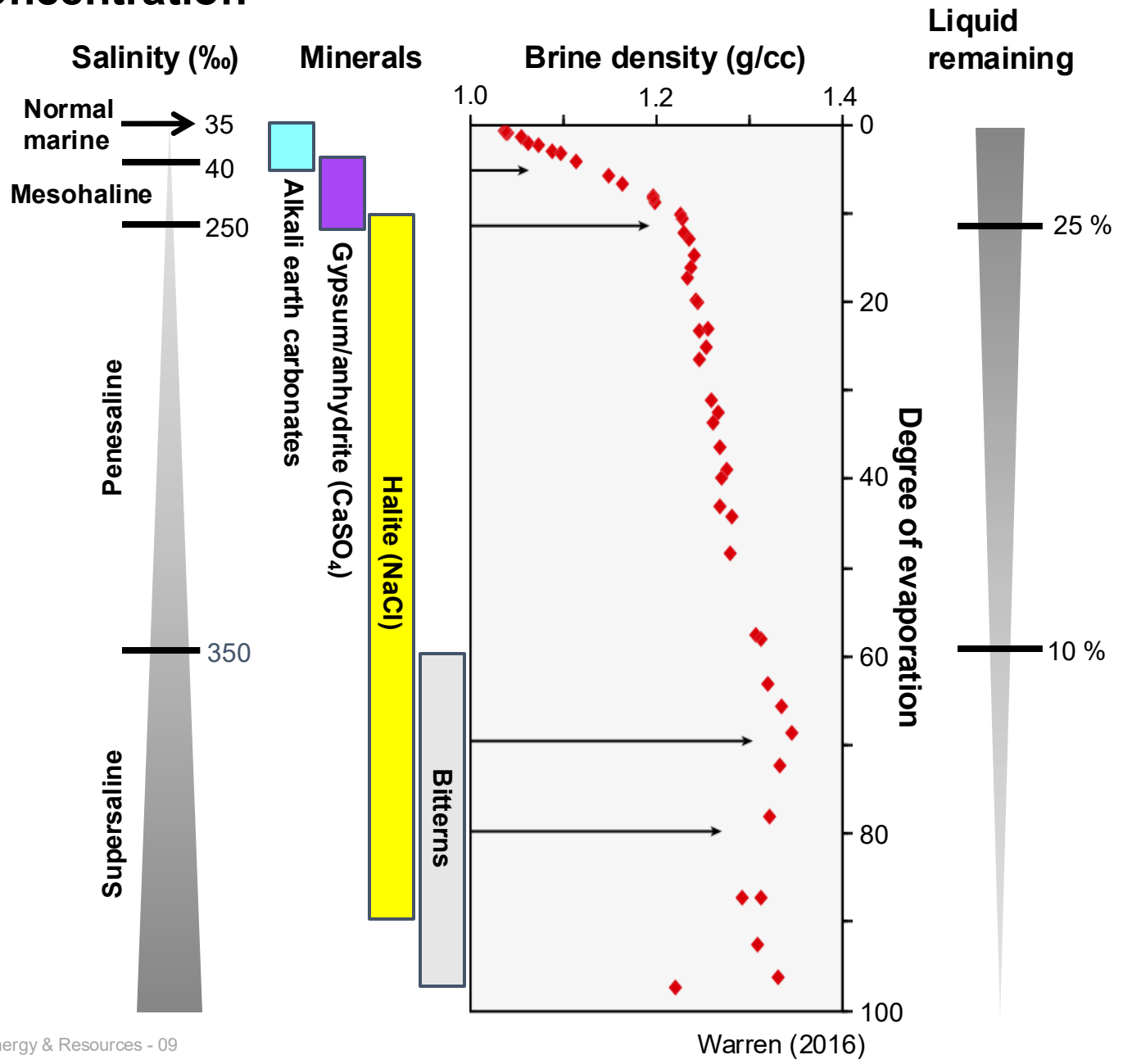


Warren (2016)

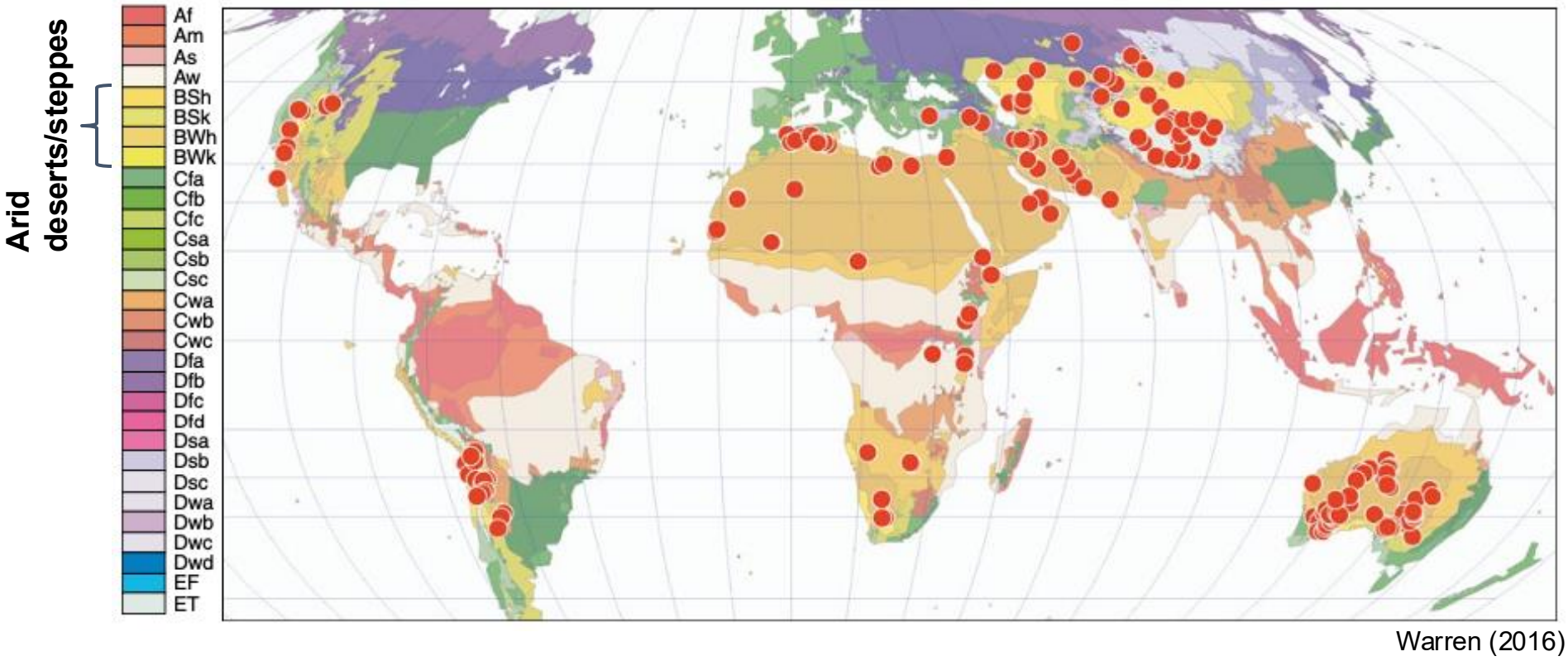
- **Evaporite minerals form from saturated brines by**
  - **solar evaporation**
  - **diagenetic processes**
- **Brine = water with dissolved salts**

Controls on evaporite formation

Brine concentration



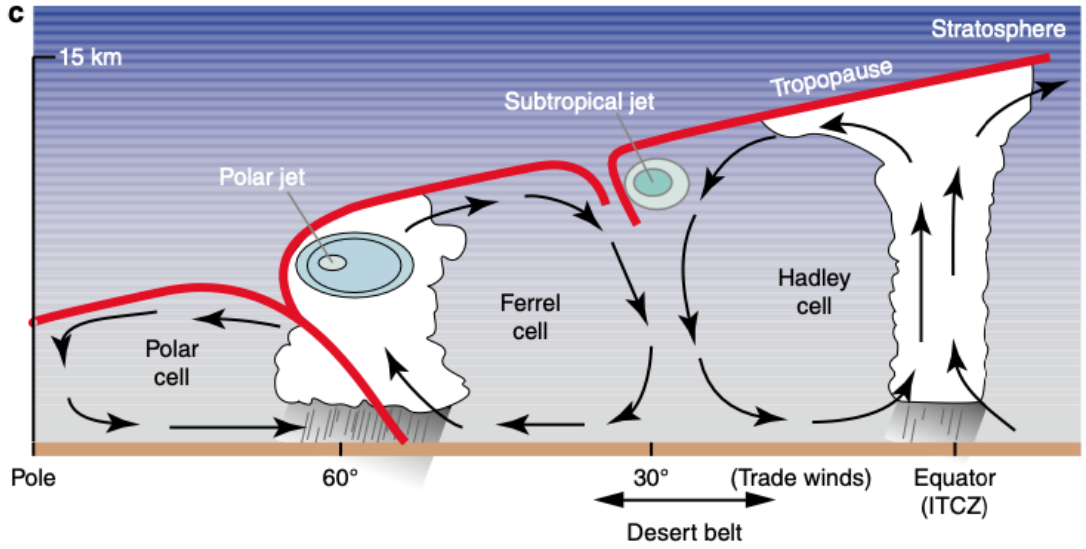
Controls on evaporite formation
Climate



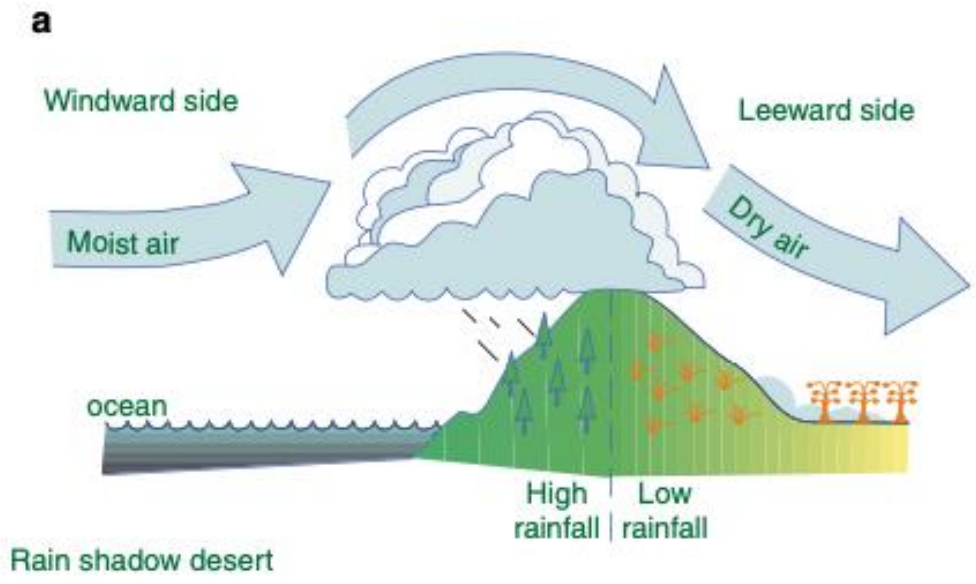
Warren (2016)

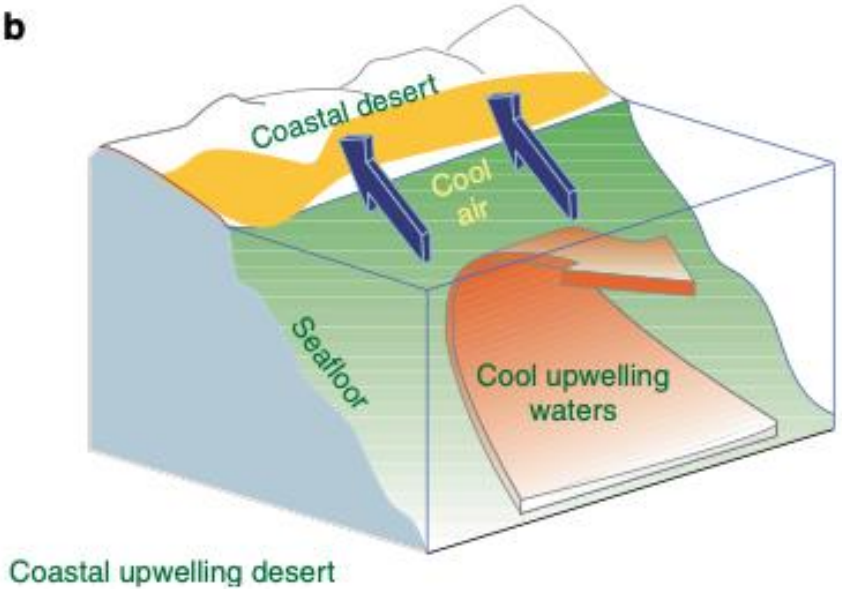
- **Dry, hot conditions favor evaporation process (deserts)**

Controls on evaporite formation
Climate



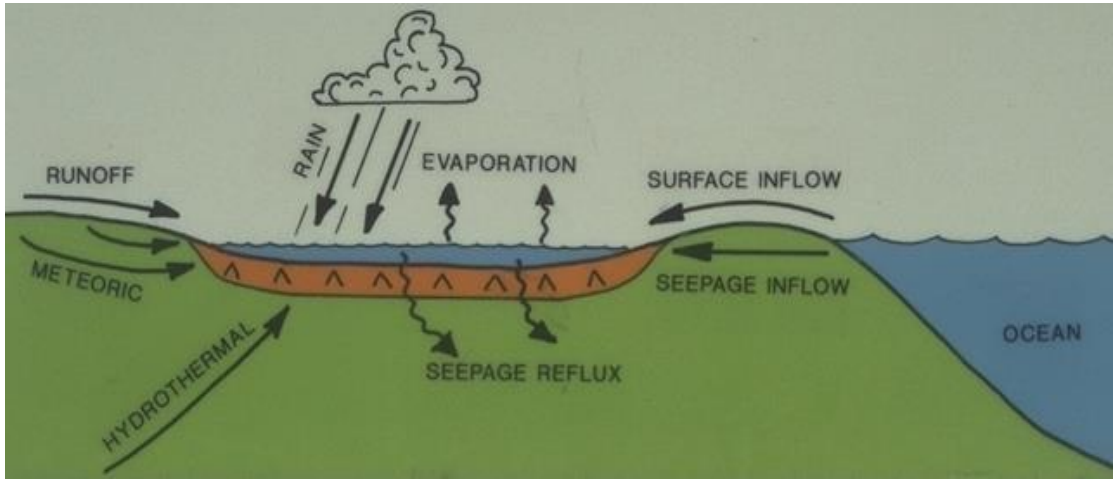
- **Distribution of dry, hot areas is determined by global atmospheric circulation and local conditions (eg rain shadow)**

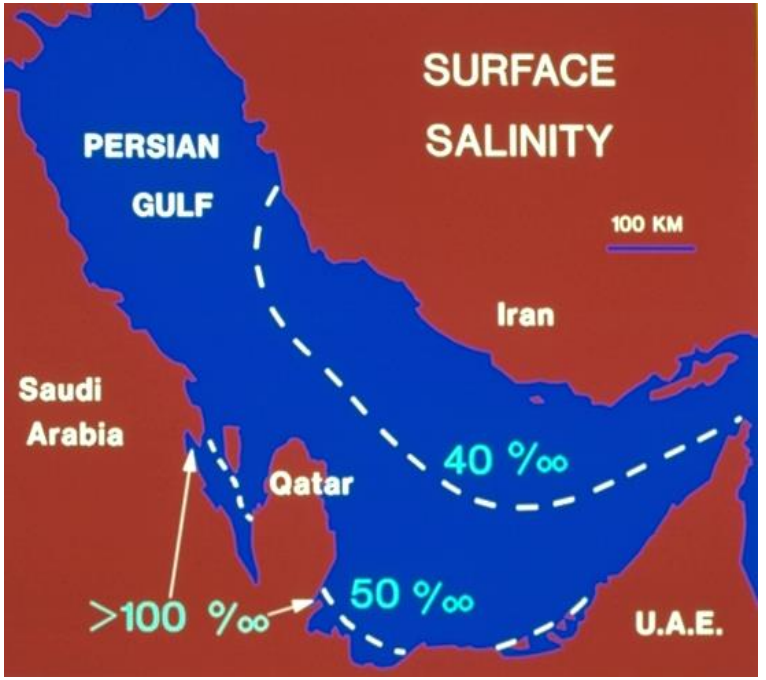




Warren (2016)

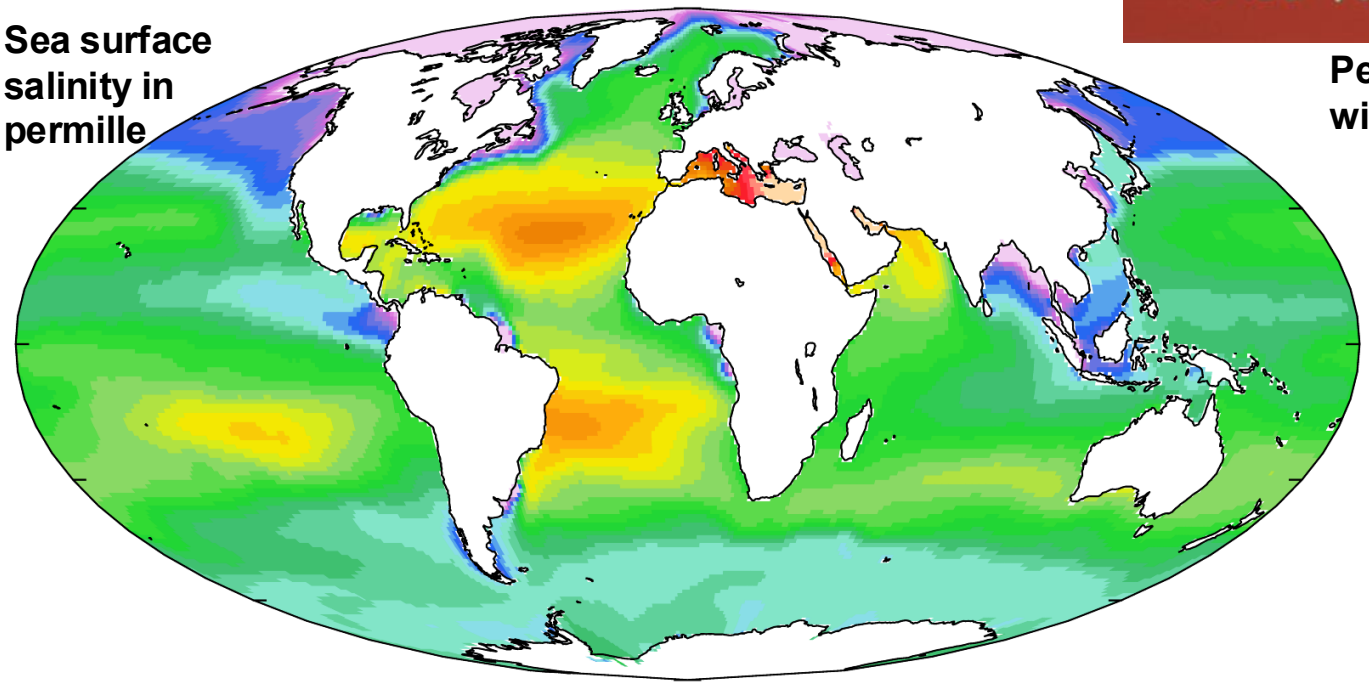
Controls on evaporite formation
Hydrology and restriction





Persian Gulf: Salinity changes with distance from inlet

Sea surface salinity in permille



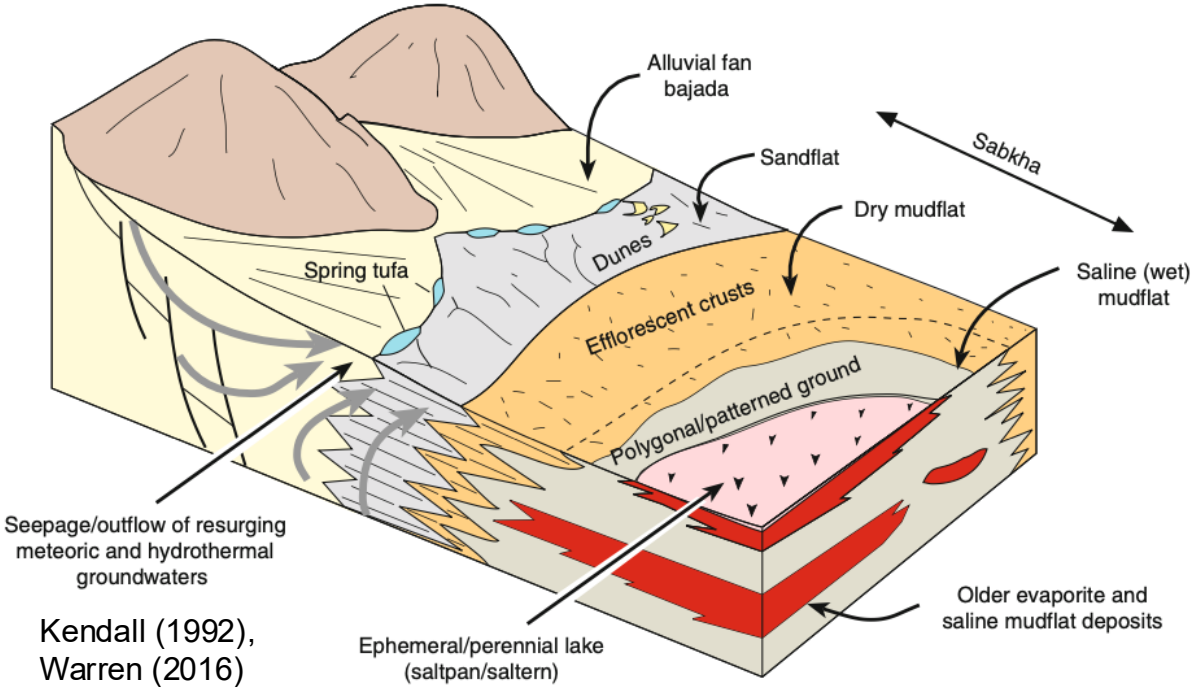
Source: World Ocean Atlas, National Oceanographic Data Center (USA)

Depositional environments
Continental basins







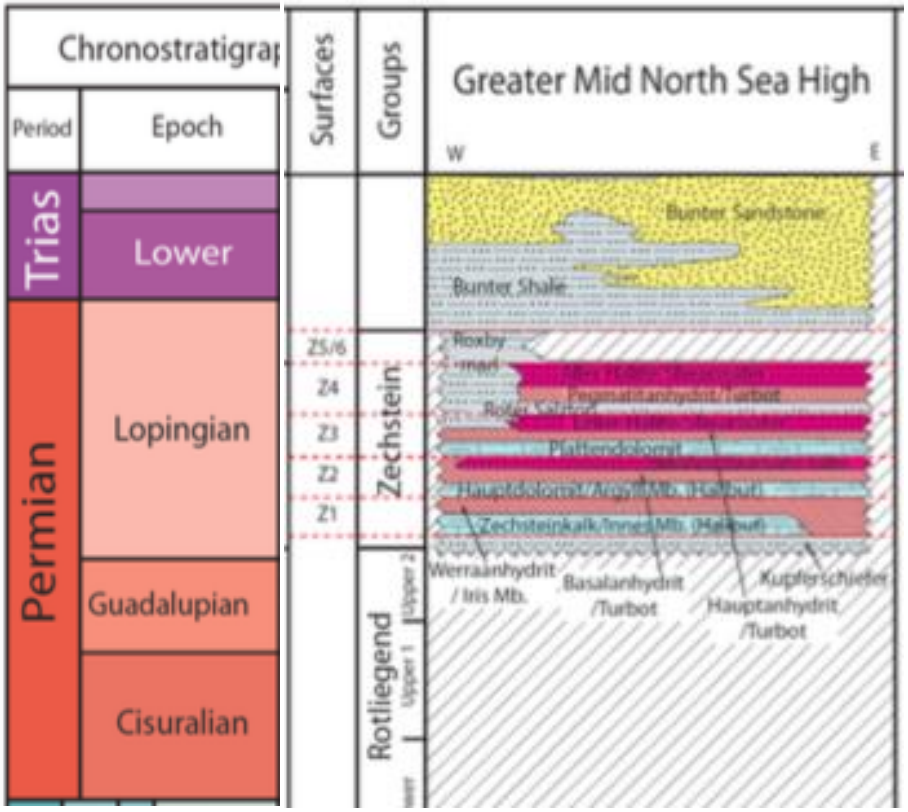


Kendall (1992), Warren (2016)

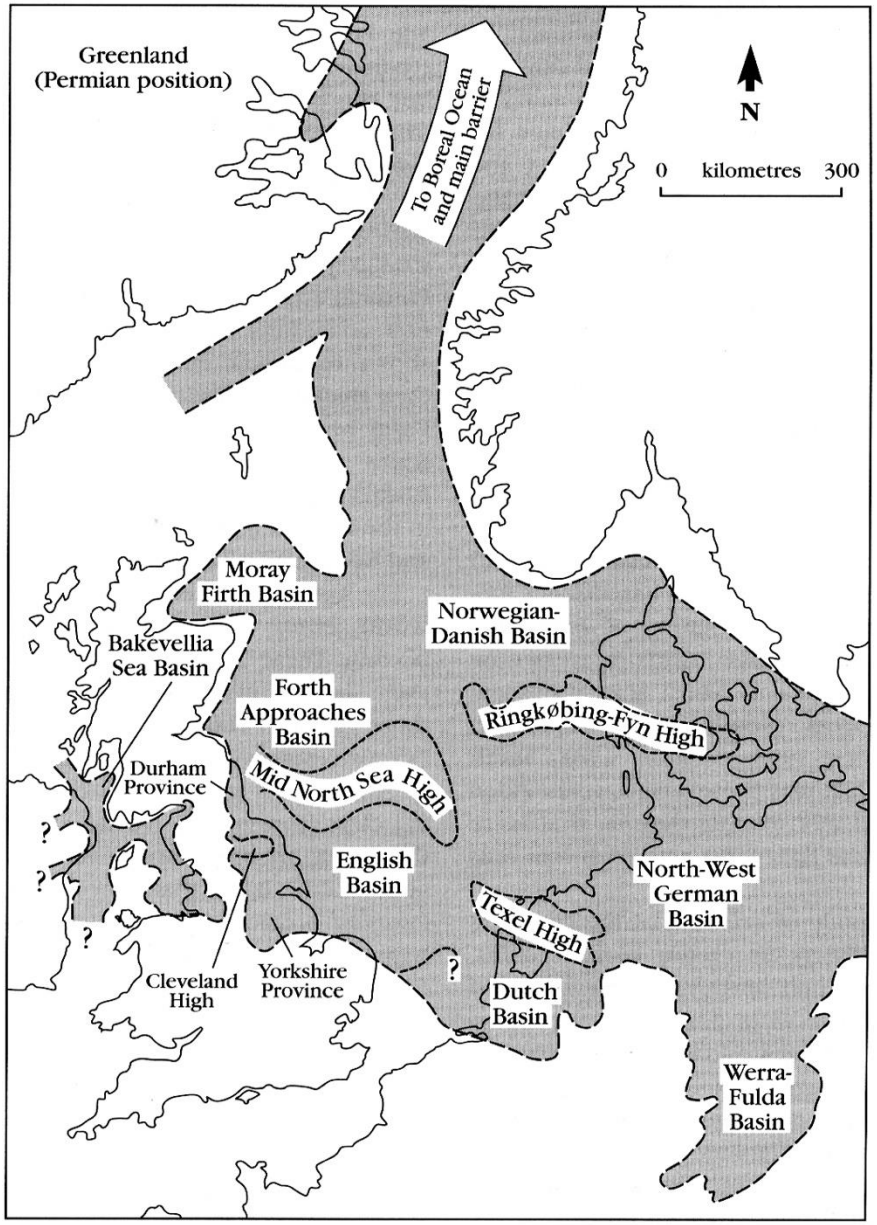
- Modern/Quaternary systems limited to coastal areas and closed continental basins
- Tectonics, hydrology and topography provide isolation and aridity for evaporation
- Evaporation of pore water – evaporite mineral growth in pre-existing sediment
- Precipitation from shallow brine pools/lakes
- Mineralogy varies with brine source and composition

Depositional environments
Marine evaporite basins – eg Zechstein

- Upper Permian carbonate-evaporite cycles reflect freshening and restriction in paleo North Sea
- Isolation and evaporation of entire basin
- Thick halite in basin center, surrounded by carbonates and sulfates



Peryt (2010)

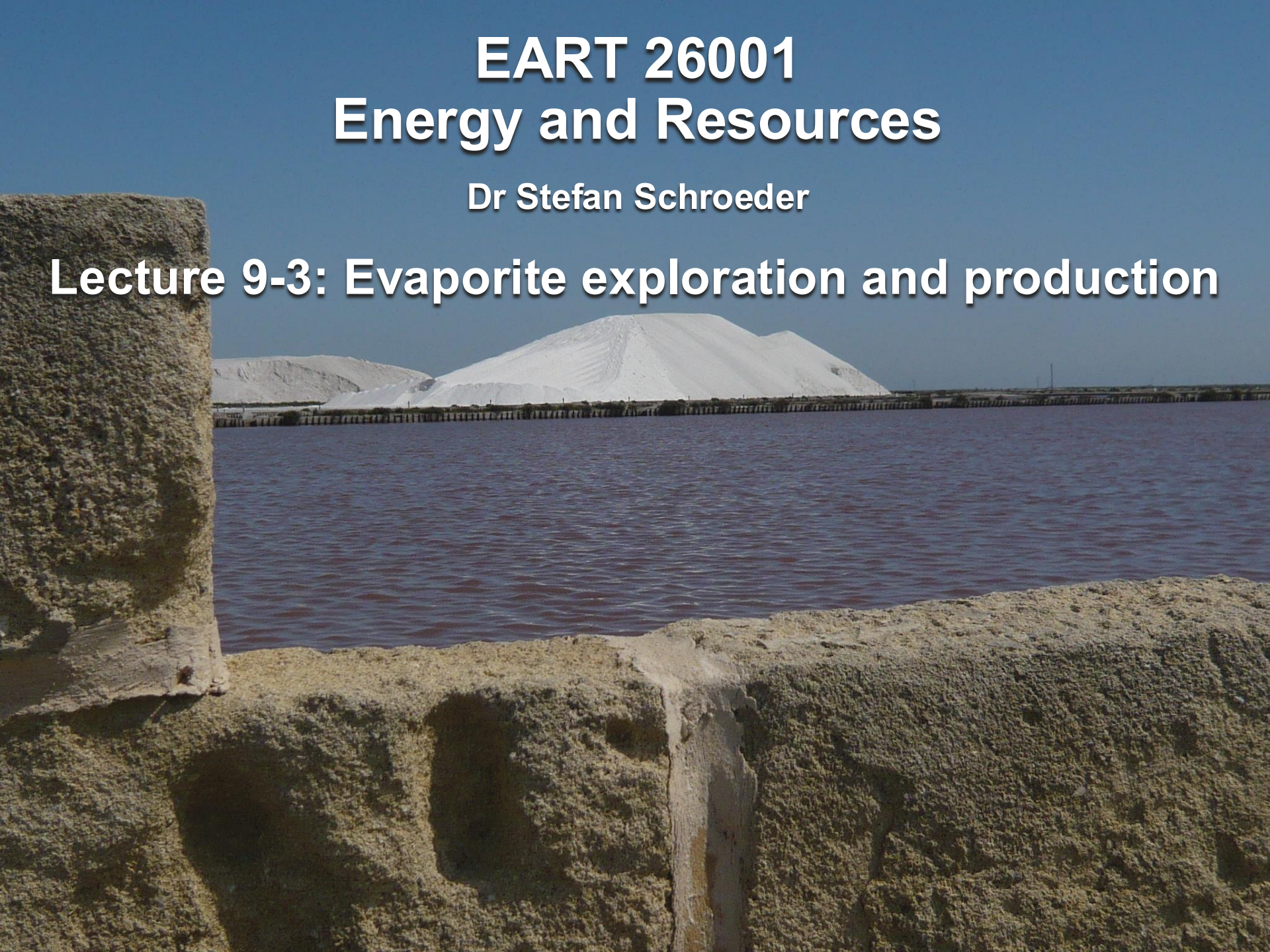


Smith (1995)

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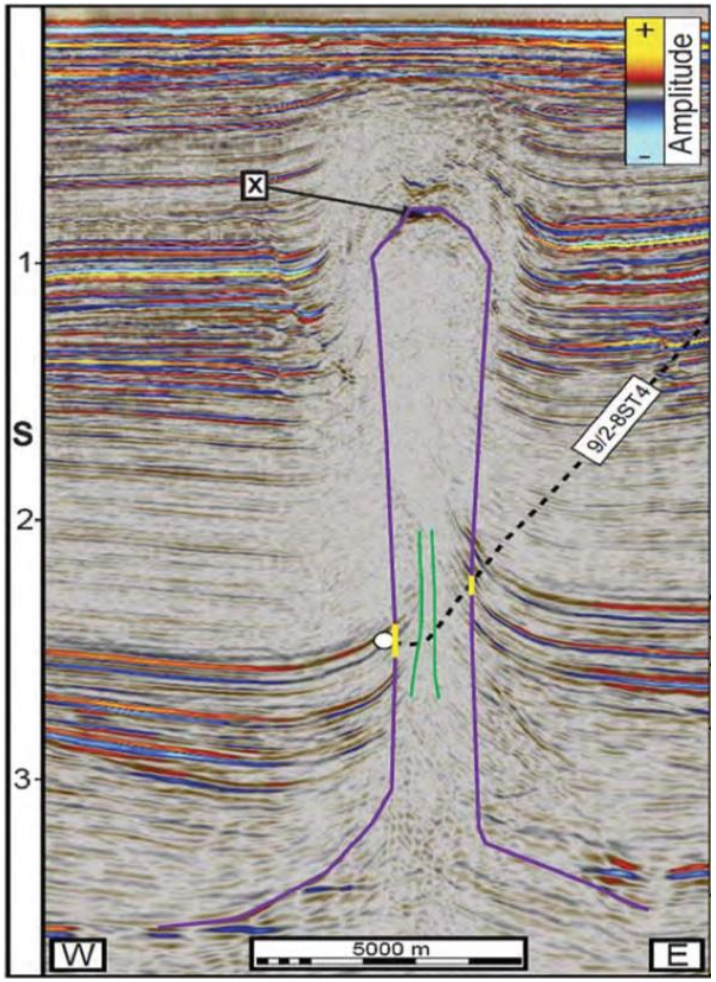
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Lecture 9-3: Evaporite exploration and production

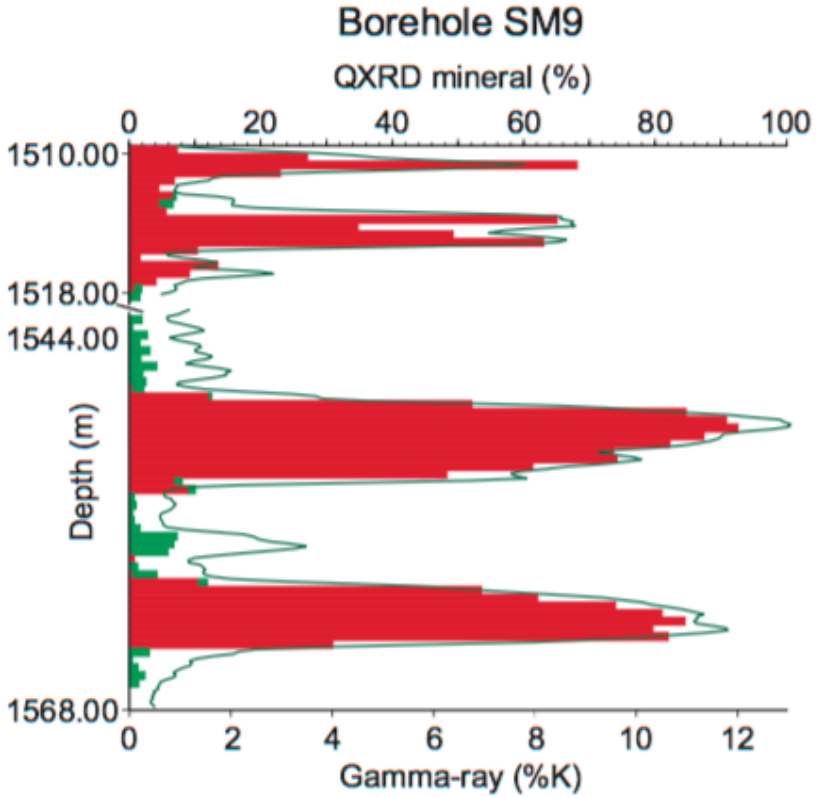


Evaporite exploration

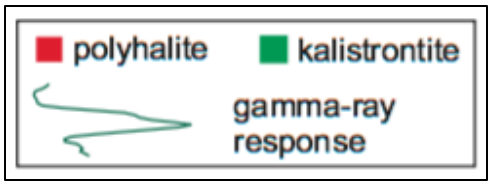
- Regional depositional and climatic history to identify times and location of evaporite formation
- Seismic
- Wireline logging (e.g. gamma-ray, density)
- Borehole geochemistry to identify mineralogy



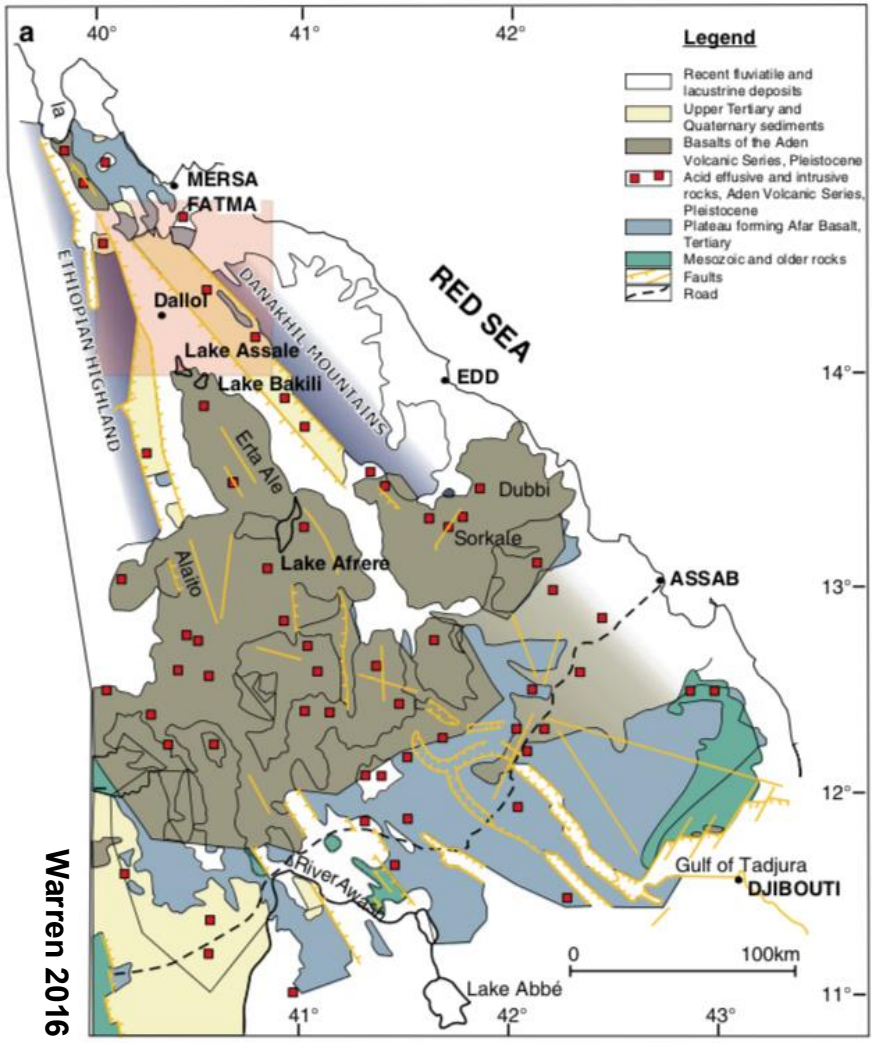
Jones 2014



Kemp 2016



Evaporite production
Mining of surface deposits



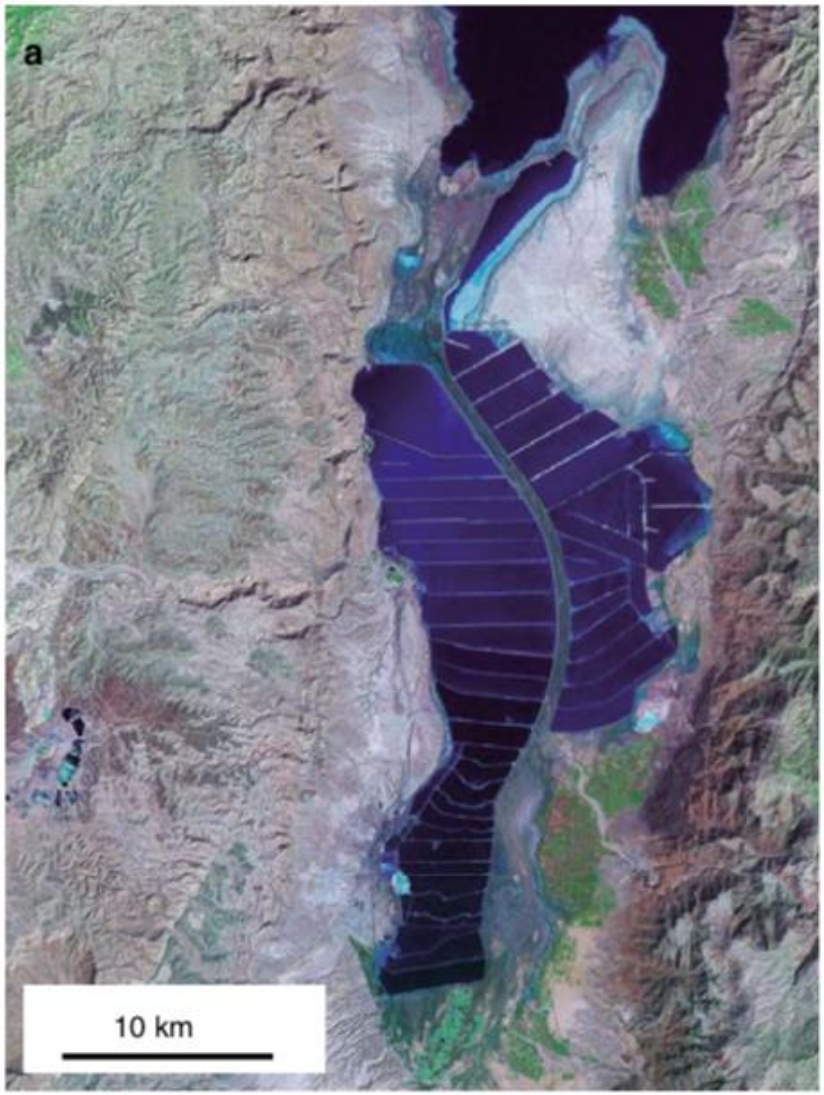
Warren 2016

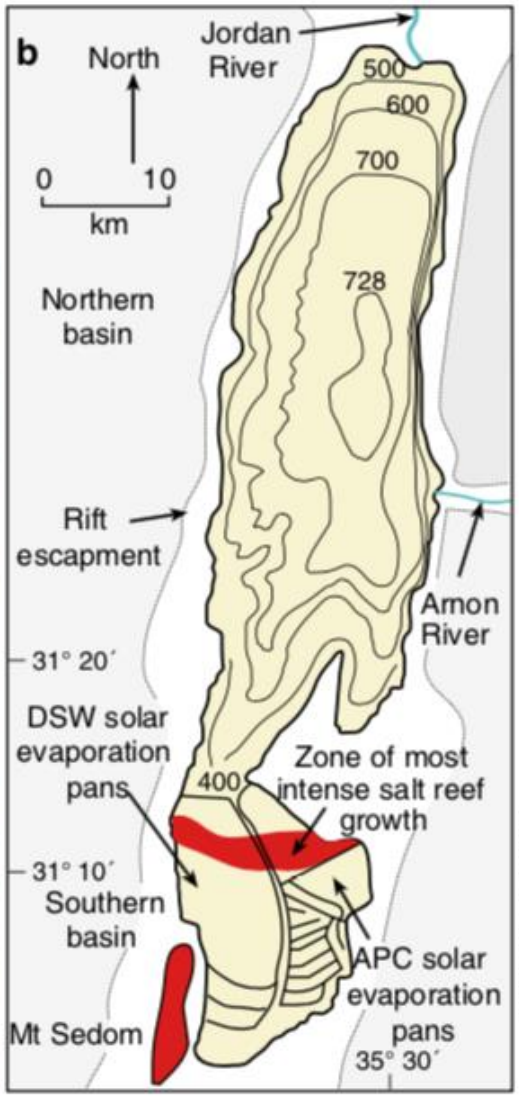


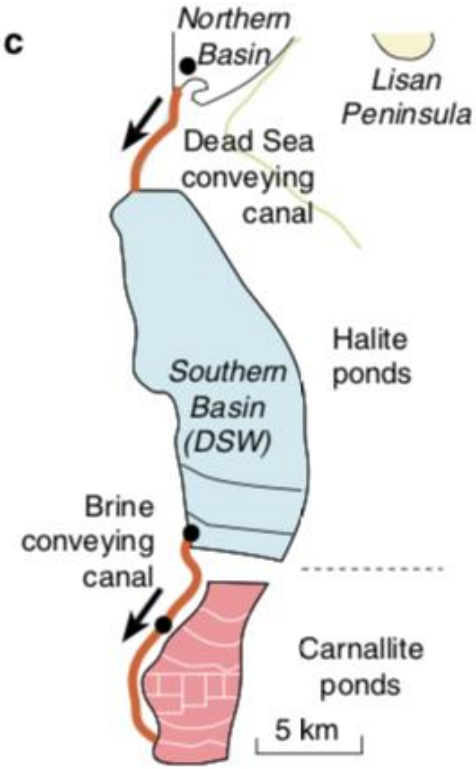


© Lafforgue/Getty images

Evaporite production
Industrial salars

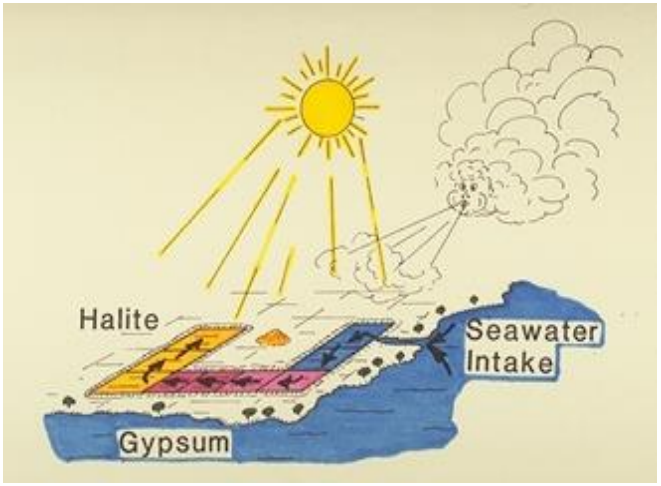


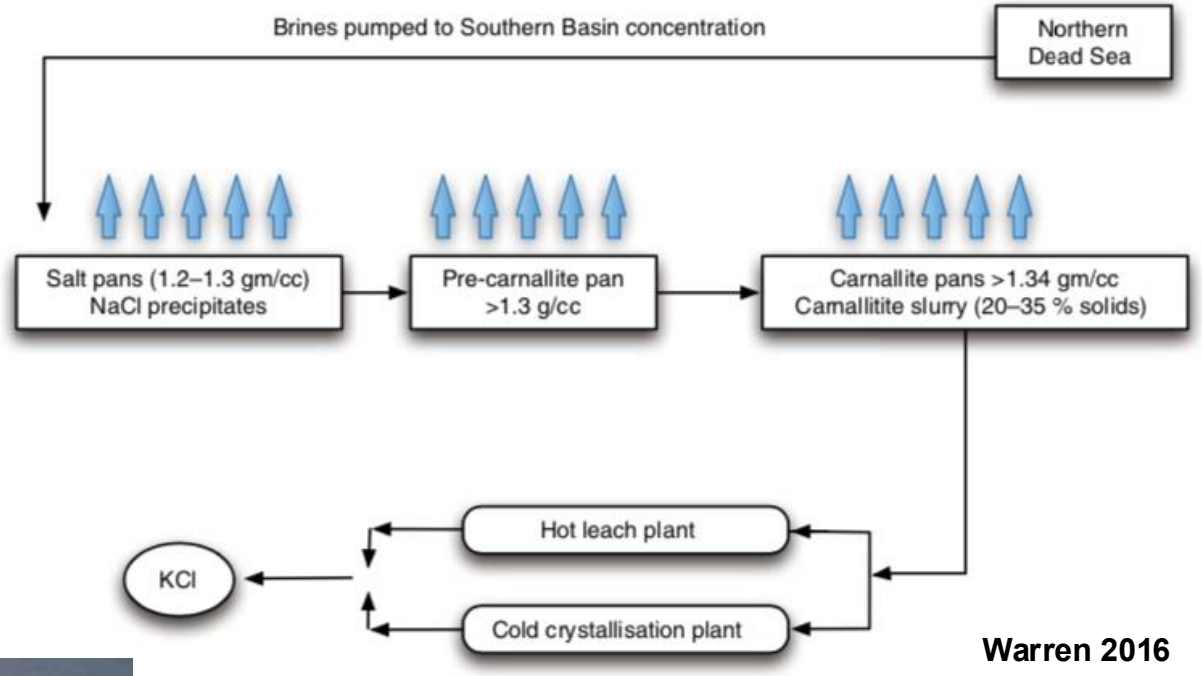




Warren 2016

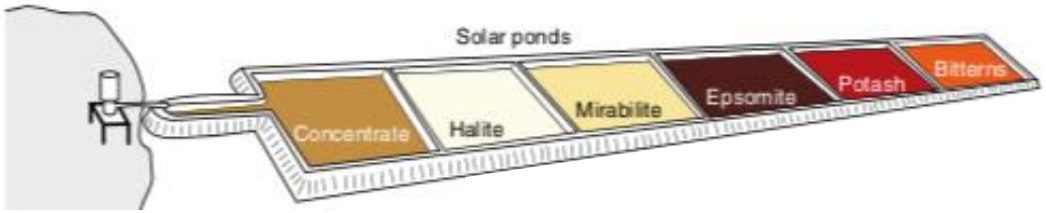
Evaporite production
Industrial salars





Warren 2016



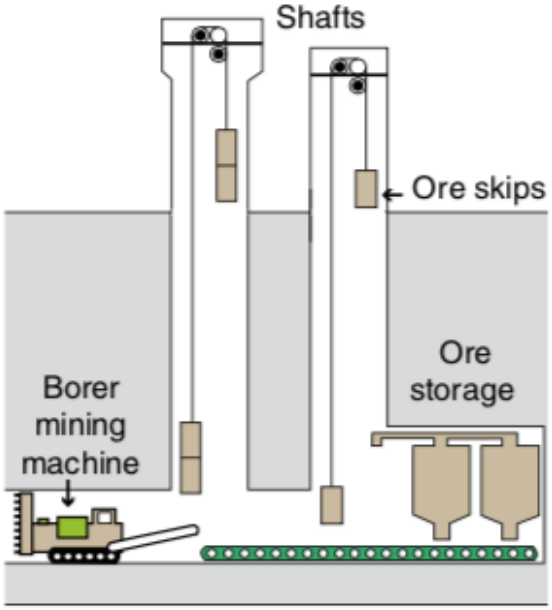


- Simple process
- High operational costs
- Large footprint with waste heaps and usage of large volumes of water

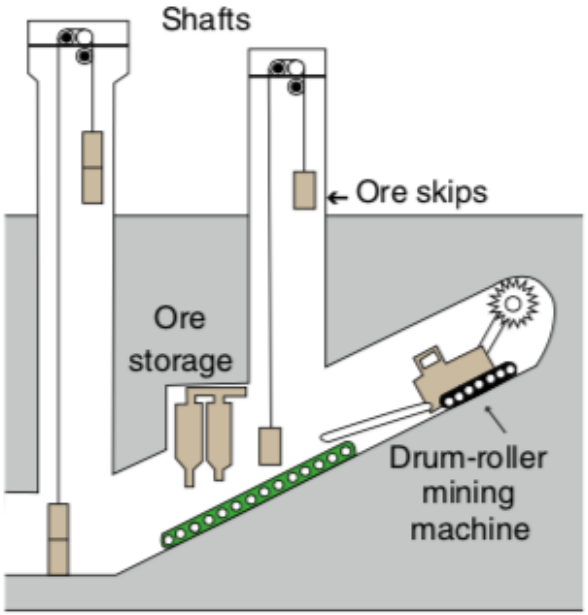
Evaporite production
Subsurface mining

- High cost for installation
- Low operational costs
- Small footprint, waste heaps





Continuous mining



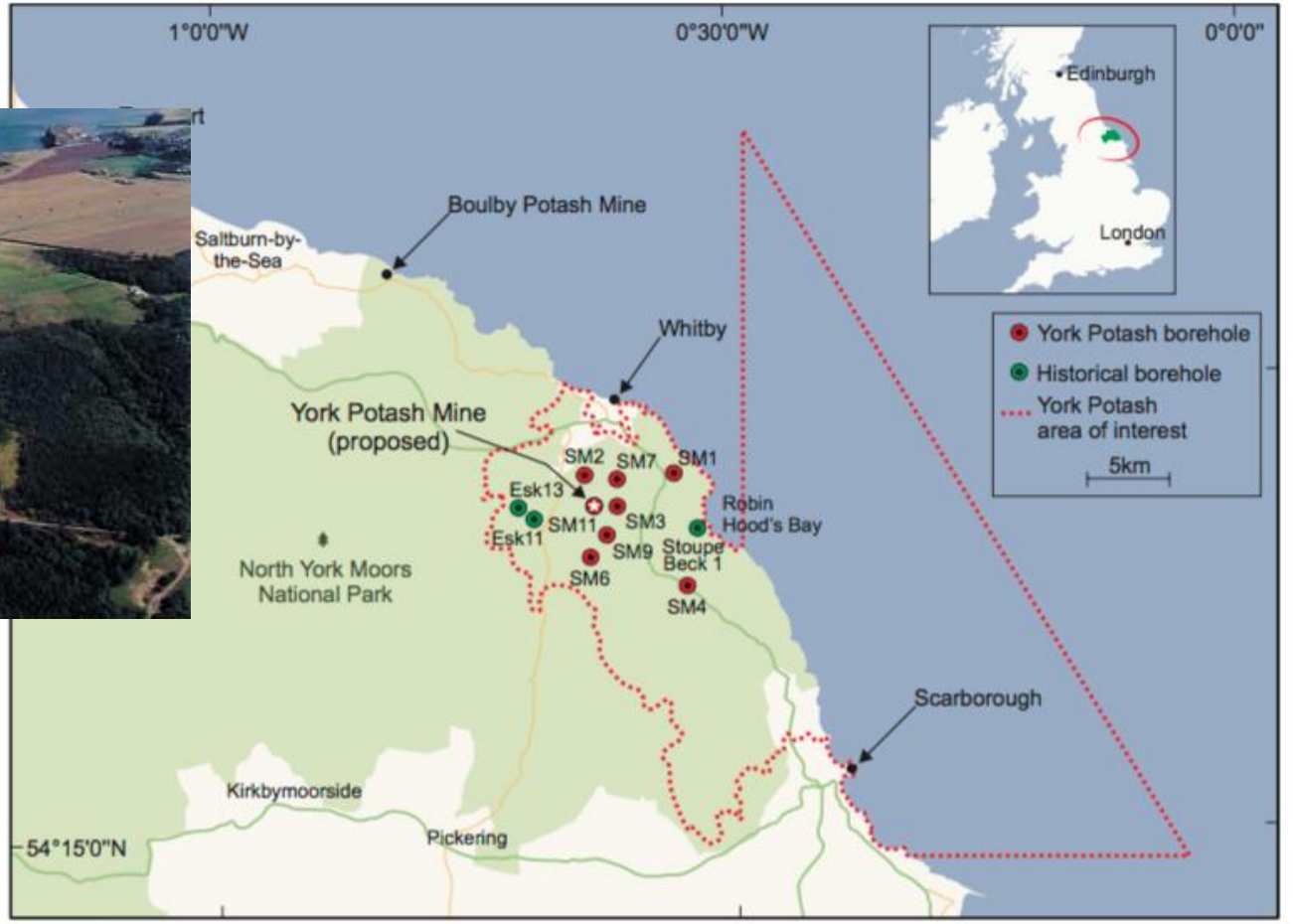
Cut & stope mining



Warren 2016

Evaporite production
Subsurface mining





Kemp 2016

- **Boulby mine, N Yorkshire**
- **Halite, potash (sylvite) and polyhalite (K_2 Ca_2 Mg $(SO_4)_4$ $2H_2O$)**
- **Depth 1.4 km, incl scientific lab at 1.1 km (study of extremophile life)**
- **Operations under sea bed**

https://youtu.be/TkMN3XlIl8A?si=rbV6ZWLCdlEv9B76

Evaporites as subsurface storage



von Berlepsch 2017

- **Documents**
- **Industrial and nuclear waste, munitions**
- **CO_2 (CCS)**
- **Liquid hydrocarbons (emergency reserves)**

Evaporite production
Environmental aspects

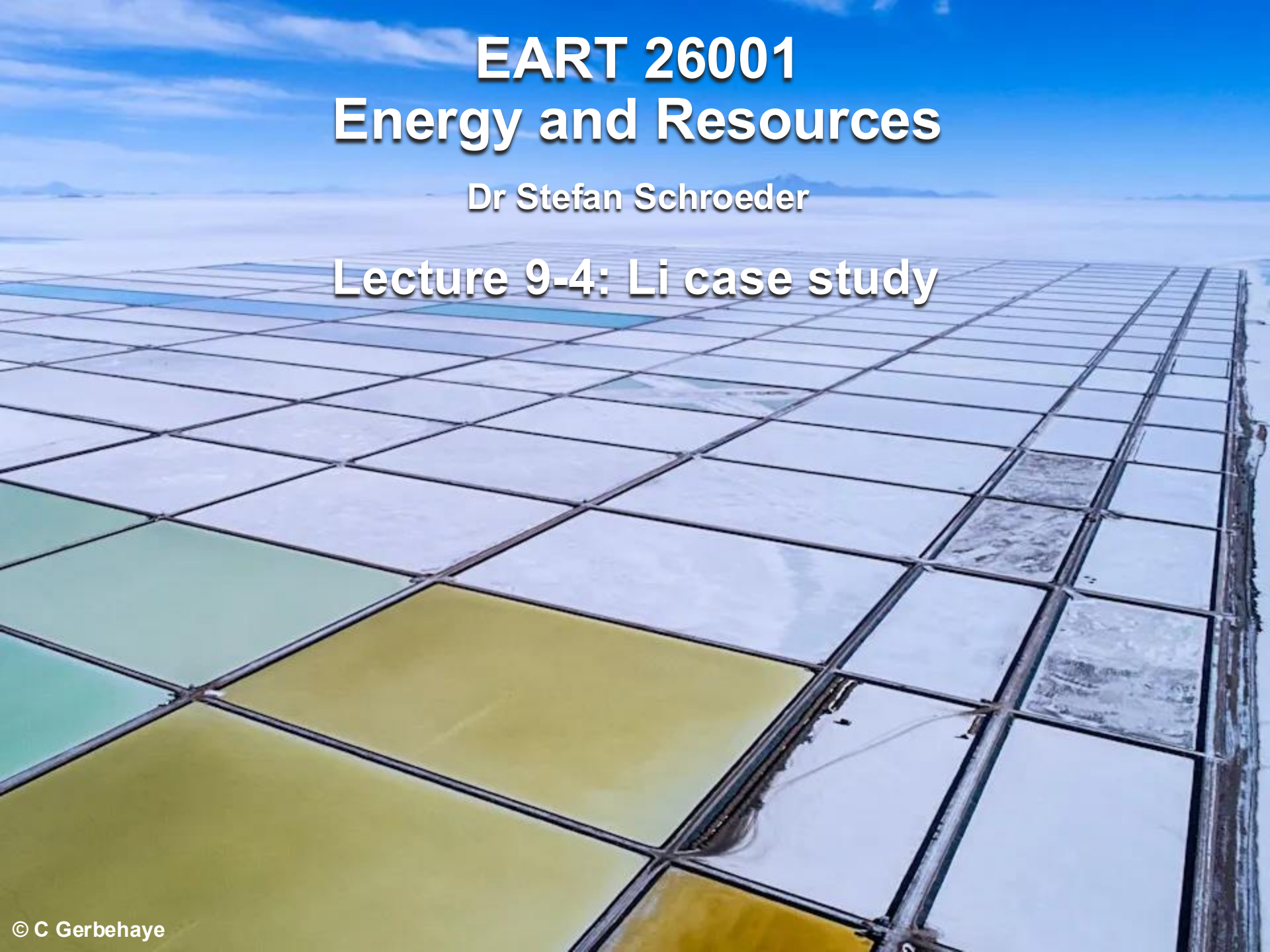
- High water use for mining in areas of limited water resource
- Leakage of processing chemicals from ponds, or of brine into groundwater
- Strip mining
- Sinkholes



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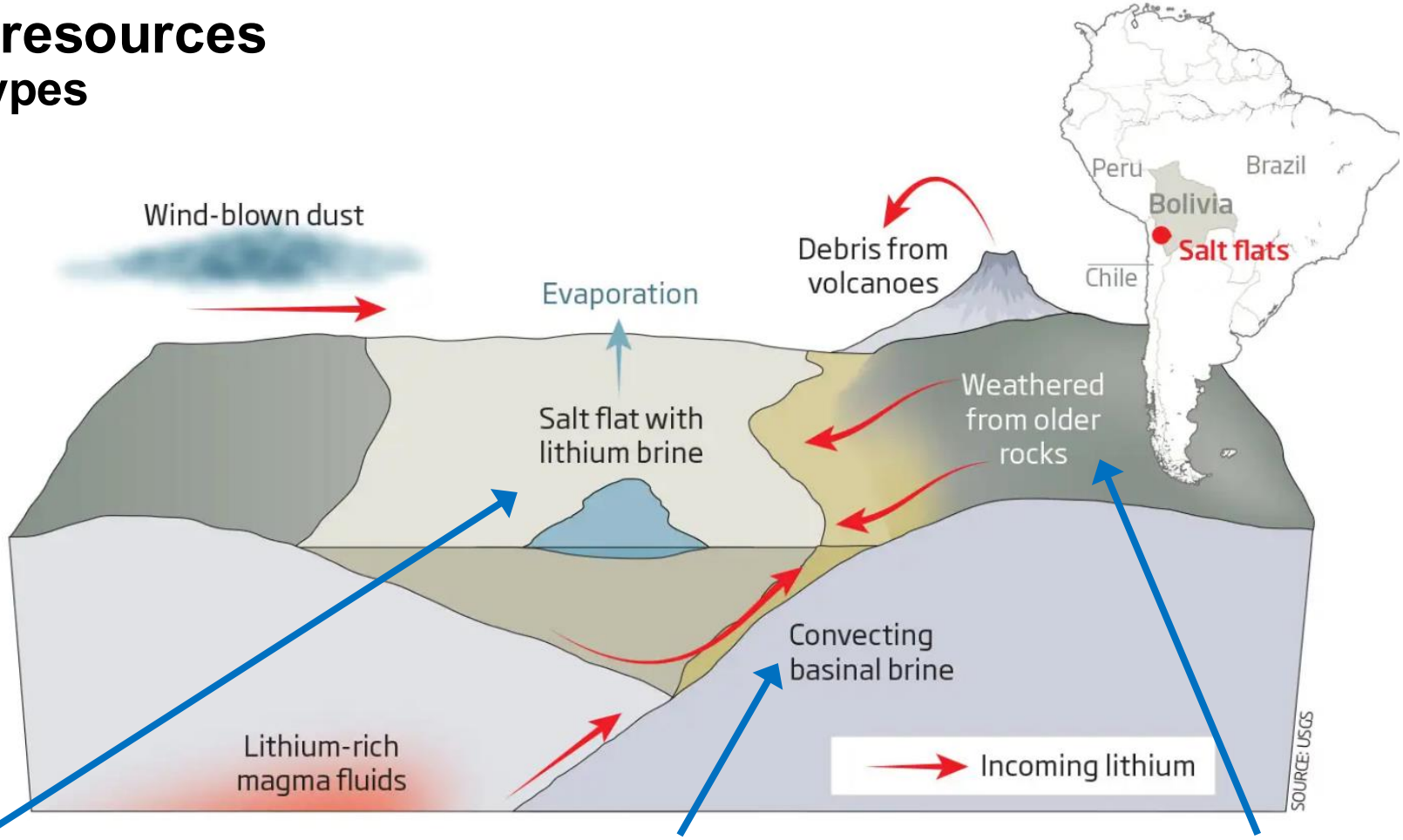
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Lecture 9-4: Li case study



© C Gerbehaye

Lithium resources
Deposit types

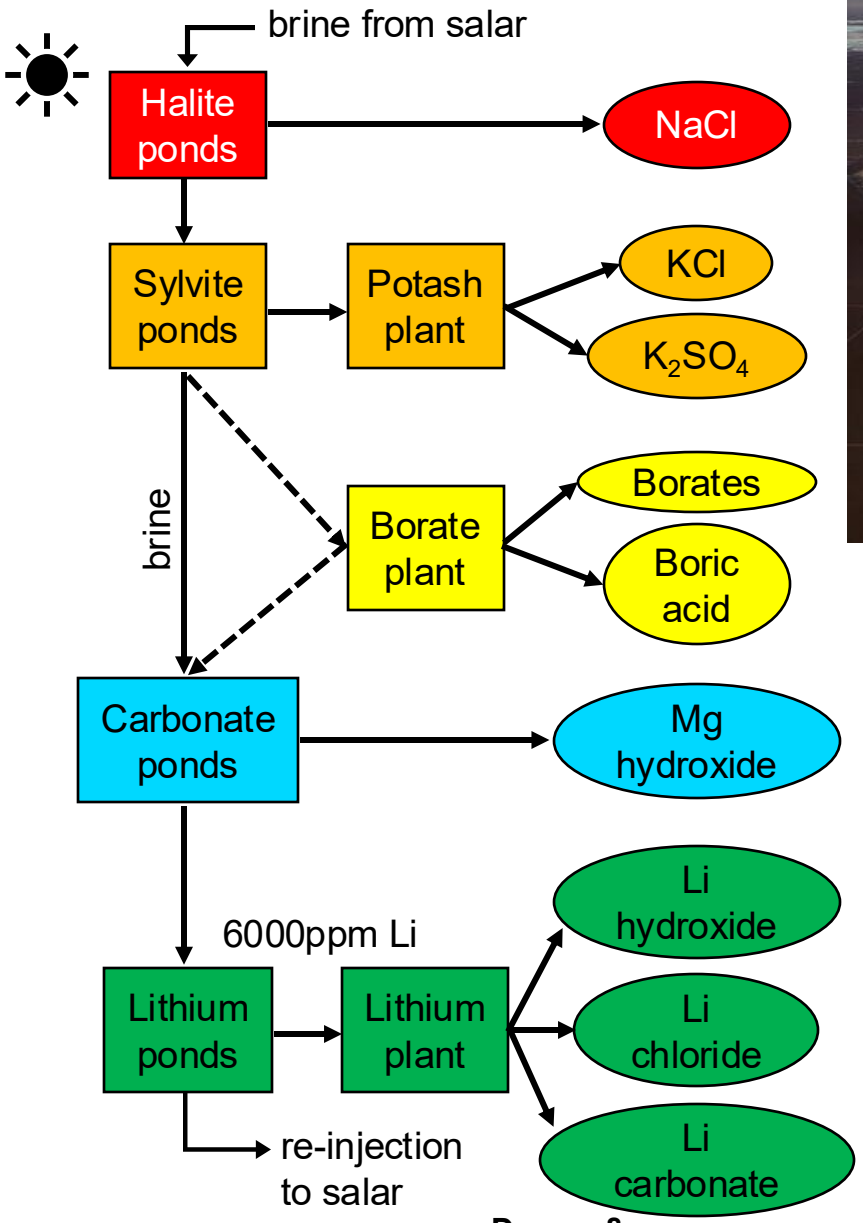


- **Brine deposits: mining of natural and artificial brine pools**
- **66% of global reserves**
- **Easy and cheap mining, but relies on water and occurs in areas with water scarcity**

- **Li in subsurface waters**
- **New techniques for Li extraction**

- **Hard rock deposits: Li-bearing minerals in igneous rocks (lithophile behavior, pegmatites)**
- **25% of global production**
- **Open pit mining: costly and time-intensive due to low concentrations**

Lithium resources
Brine deposits



Draper & Gerbehaye (2019)

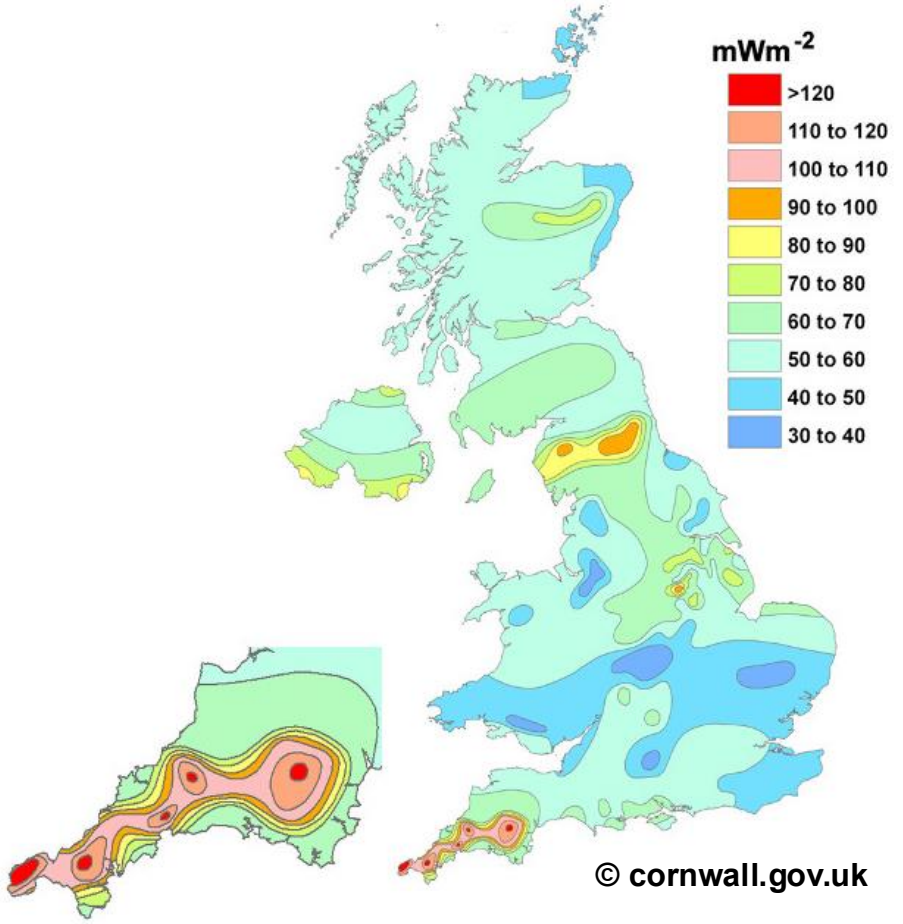


© Reuters/Ivan Alvarado

- **Mainly Lithium triangle Argentina-Bolivia-Chile and Lithium valley in W Australia**
- **Example Atacama: Mud flats and small pools in north allow pre-concentration of brine and chloride precipitation in S**
- **Li sources**
  - **Geyser field: geothermal fluids 28-47ppm Li**
  - **Subsurface brines and intermittent streams on flanks**
- **Extraction as Li-chloride salt, average Li concentration 1500ppm**
- **Immense water consumption in arid area**

Lithium resources
Hard rock deposits





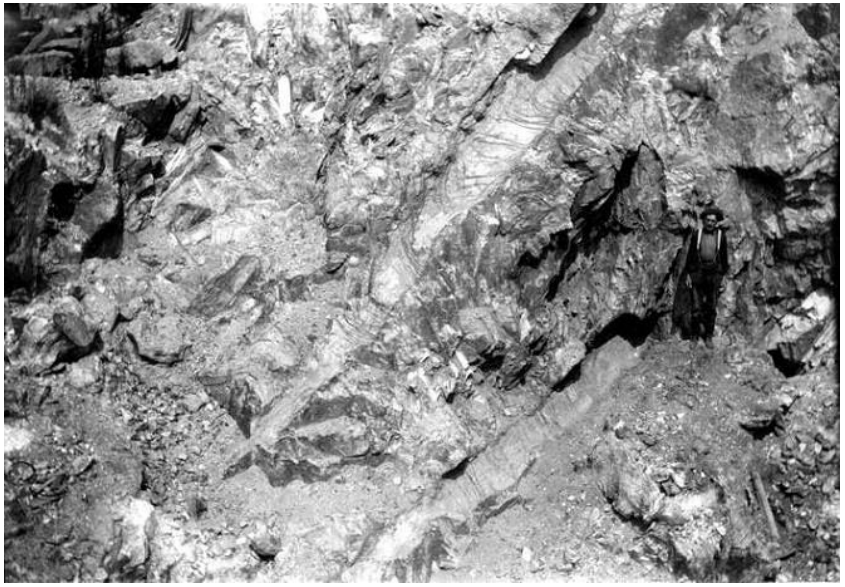
- Li-enriched hydrothermal waters were noticed in 18th century Cu and Sn mines of Cornwall, but considered ‘useless’ until now
- Li associated with High-heat producing (HHP) granites: geothermal energy projects (see Uranium lecture)
- Li minerals crystallize in/around large igneous intrusions (lithophile element)

Lithium resources
Hard rock deposits

- **Minerals:**
  - **Spodumene: $LiAlSi_2O_6$**
  - **Poly/Tri-lithionite (old names lepidolite and zinnwaldite): $KLi_2Al(Si_4O_{10})(F,OH)_2$**
- **Typically associated with pegmatites:**
  - **coarse-grained rocks**
  - **crystallized in/around igneous intrusions (mostly granitic)**
  - **likely involve magmatic-hydrothermal fluids**
- **Weathering of Li-rich micas produces Li-enriched clay minerals**



Poly/Tri-lithionite (Lepidolite, Li mica)
© Volker Betz (2012)



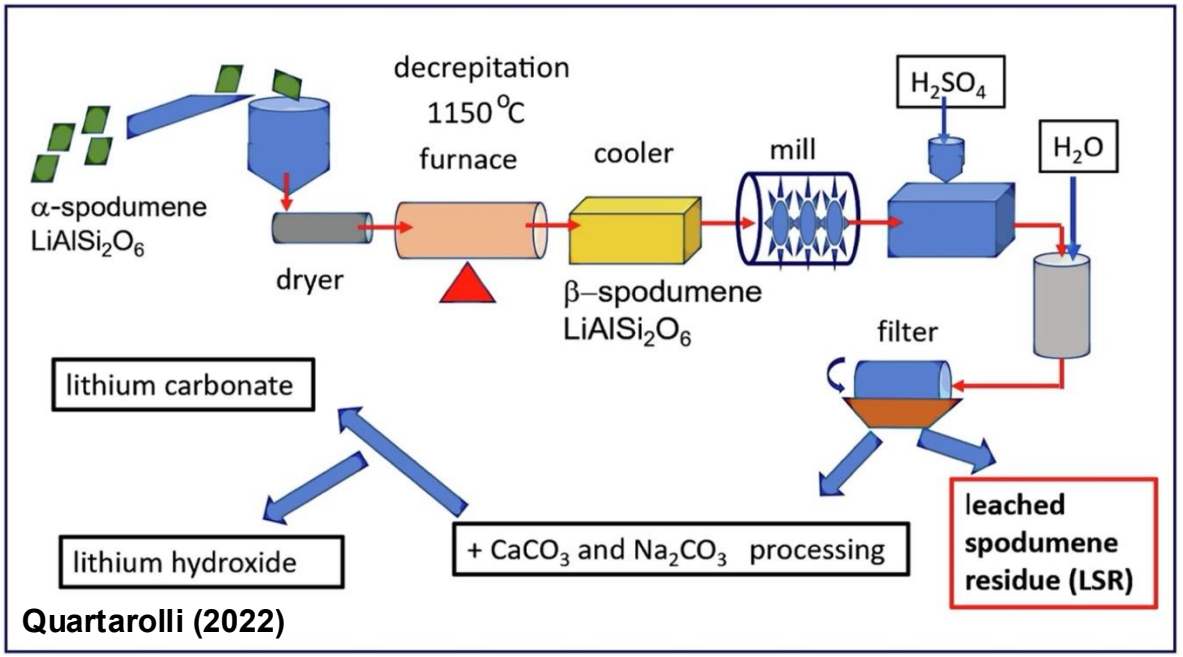
m-scale spodumene crystals in pegmatite
© Thomas Krassmann

Lithium resources
Hard rock deposits

- Trelavour Downs (Cornwall): exploration in kaolinite clay (China clay) rich in Li micas
- Grade: 0.25% Li oxide
- Ore processing: concentration of Li mica, followed by extraction of battery-grade Li hydroxide via acid leaching
- Spodumene processing involves roasting and carbonate leaching to produce Li carbonate
- High C footprint of processing

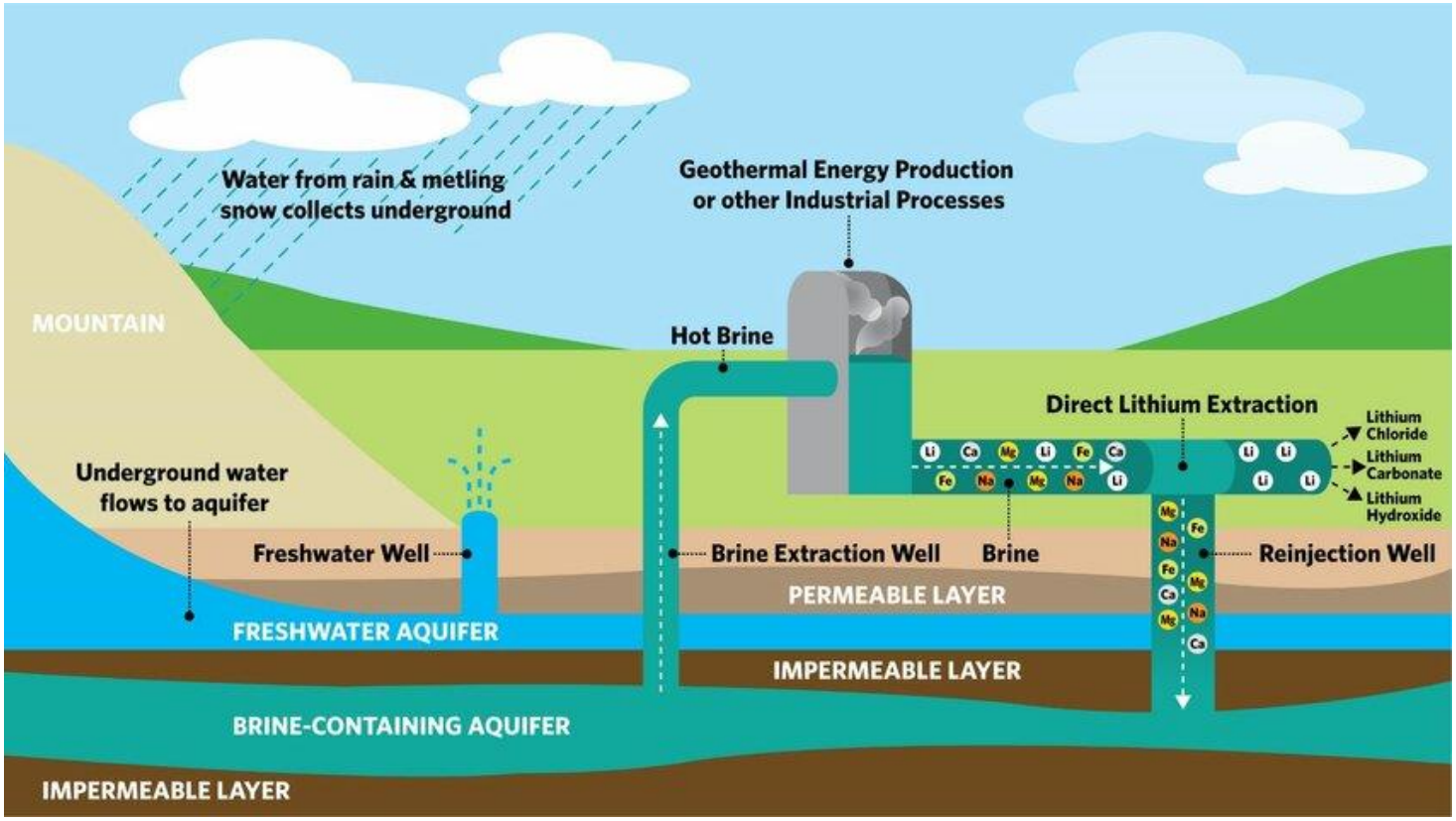


© Cornish Lithium



Quartarolli (2022)

Lithium resources
Li from subsurface waters



Parker (2022)

- Combined geothermal heat production and Li extraction from geo/hydrothermal waters (<220ppm Li in Cornwall)
- How to extract just Li?
- Direct Lithium Extraction (DLE) extracts lithium chloride which is then purified and concentrated to produce lithium hydroxide

Evaporite deposits
Summary week 9

- Evaporites form when brines become oversaturated with ions, because of solar evaporation or diagenetic processes. The mineralogy is highly variable and reflects composition of the brine (and thus its source terrains).
- The mineralogical variability translates into a range of applications. Key raw materials deriving from evaporation processes are potassium and lithium.
- Formation and preservation require arid climate, negative water balance, and restriction from source of brine. Evaporite minerals typically form in playas, sabkhas, and salinas (saline lakes). Thick and cyclic evaporite successions require repeated replenishment of brine pool.
- Evaporite mining occurs via precipitation in solar ponds (salars), or subsurface mining by conventional galleries or solution mining. It usually involves a large industrial footprint and water consumption.